

datalex



What’s Big Data ?



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Data Expert
(Audit, Modeling, Urbanization, BI's project – Azure, Big Data, Spark, openLineage & Delta lover)

Trainer
(BI, Big Data, MongoDB, NoSQL, SQL, Cloud Computing)

Speaker
(Pass Summit 2018, Spark AI Tour 2020, …)

Cloud & data tech articles
(medium, LinkedIn, slideshare)

Avanade
Sr Anls, Data Analyst

2016 - 2019

Trainer
2016 - x

ESAIP, ESME, ECS, NEOMA, ESEO, EFREI, Sorbonne Paris

BI, Big Data, MongoDB, NoSQL, Panorama de la data, Cloud computing

OpenClassroom
Mentor

2019 - x

Datalex
2019 - x

Data Architect Freelance

DataRedKite
CTO & Co-Founder

2020 - 2023

DataGalaxy
2021 - x

Head of Data & AI Engineer

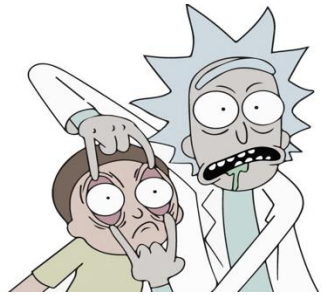
Don’t get lost



Practice



Demo

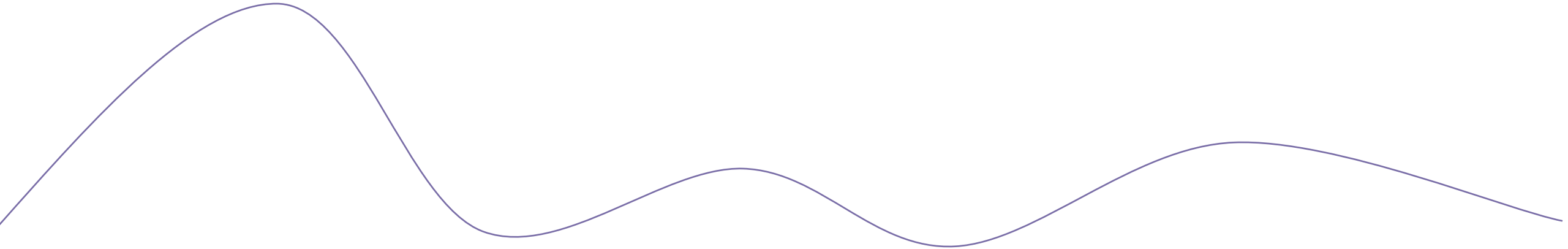


Important

Learning Objectives

1. Understand what Big Data is
2. Understand Hadoop framework
3. Understand No SQL Family
4. Understand The Data Engineering Lifecycle

BIG DATA

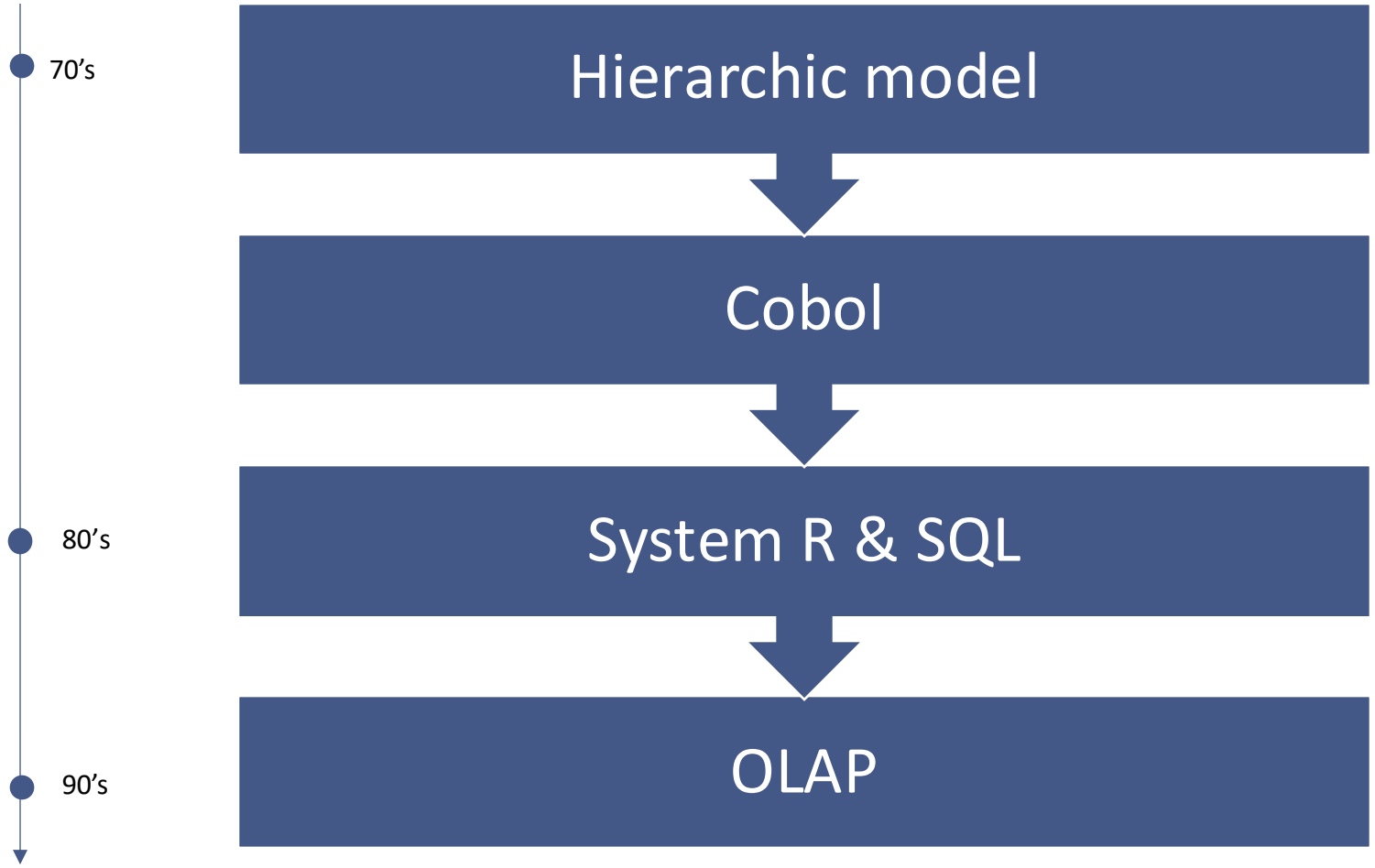


DATA

“On the most fundamental level, a database needs to do two things: when you give it some data, it should store the data, and when you ask it again later, it should give the data back to you.”

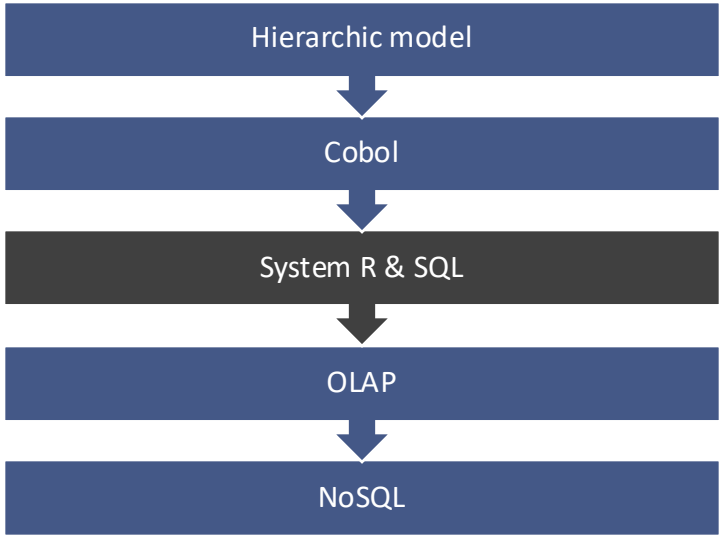
Designing Data-Intensive Applications – Martin Kleppmann

Data management



Data management

RDBMS

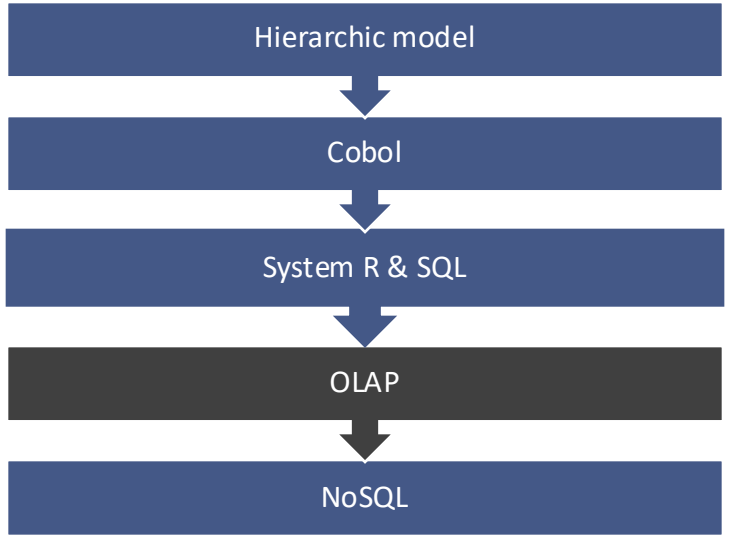


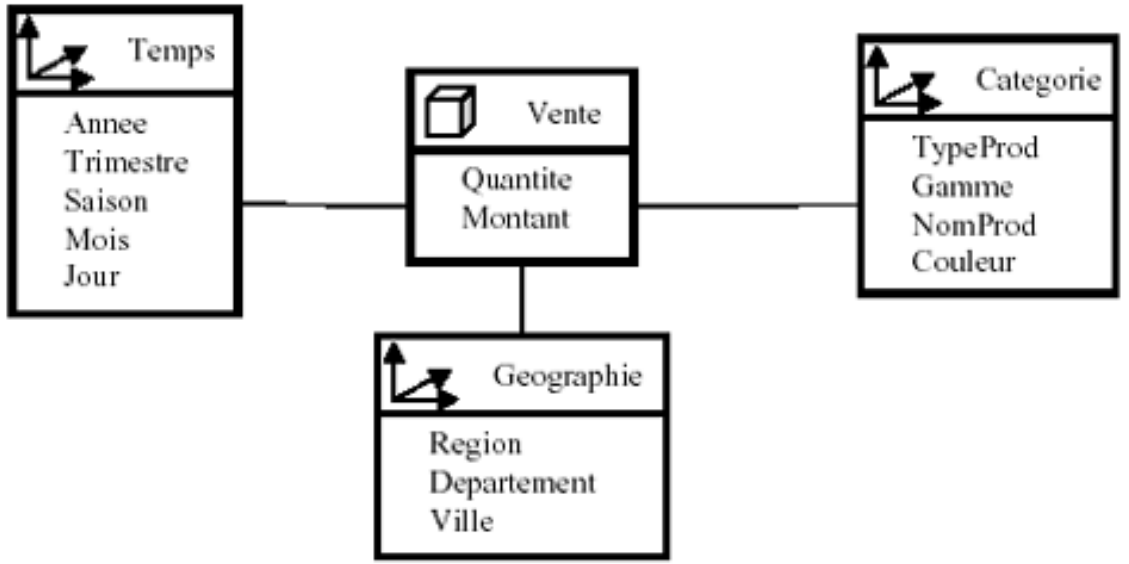
Codd's 12 Rules

- Rule 1: Information Rule
- Rule 2: Guaranteed Access Rule
- Rule 3: Systematic Treatment of NULL Values
- Rule 4: Active Online Catalog
- Rule 5: Comprehensive Data Sub-Language Rule
- Rule 6: View Updating Rule
- Rule 7: High-Level Insert, Update, and Delete Rule
- Rule 8: Physical Data Independence
- Rule 9: Logical Data Independence
- Rule 10: Integrity Independence
- Rule 11: Distribution Independence
- Rule 12: Non-Subversion Rule

Data management

OLAP : Online Analytical Processing





Some numbers

Social Media:

Every minute of the day:

- Snapchat users share 527,760 photos
- More than 120 professionals join LinkedIn
- Users watch 4,146,600 YouTube videos
- 456,000 tweets are sent on Twitter
- Instagram users post 46,740 photos

Communication:

Every minute of the day:

- We send 16 million text messages
- There are 990,000 Tinder swipes
- 156 million emails are sent; worldwide it is expected that there will be 2.9 billion email users by 2019
- 15,000 GIFs are sent via Facebook messenger
- Every minute there are 103,447,520 spam emails sent
- There are 154,200 calls on Skype

https://www.forbes.com/sites/bernardmarr/2018/05/21/how-much-data-do-we-create-every-day-the-mind-blowing-stats-everyone-should-read/#559c6dee60ba

Network traffic

We conduct more than **half of our web searches from a mobile phone** now.

More than 3.7 billion humans use the internet (that's a growth rate of **7.5 percent** over 2016).

On average, Google now processes more than 40,000 searches EVERY second (3.5 billion searches per day)!

While 77% of searches are conducted on Google, it would be remiss not to remember other search engines are also contributing to our daily data generation. Worldwide there are 5 billion searches a day.

https://www.forbes.com/sites/bernardmarr/2018/05/21/how-much-data-do-we-create-every-day-the-mind-blowing-stats-everyone-should-read/#559c6dee60ba

Devices and IOT

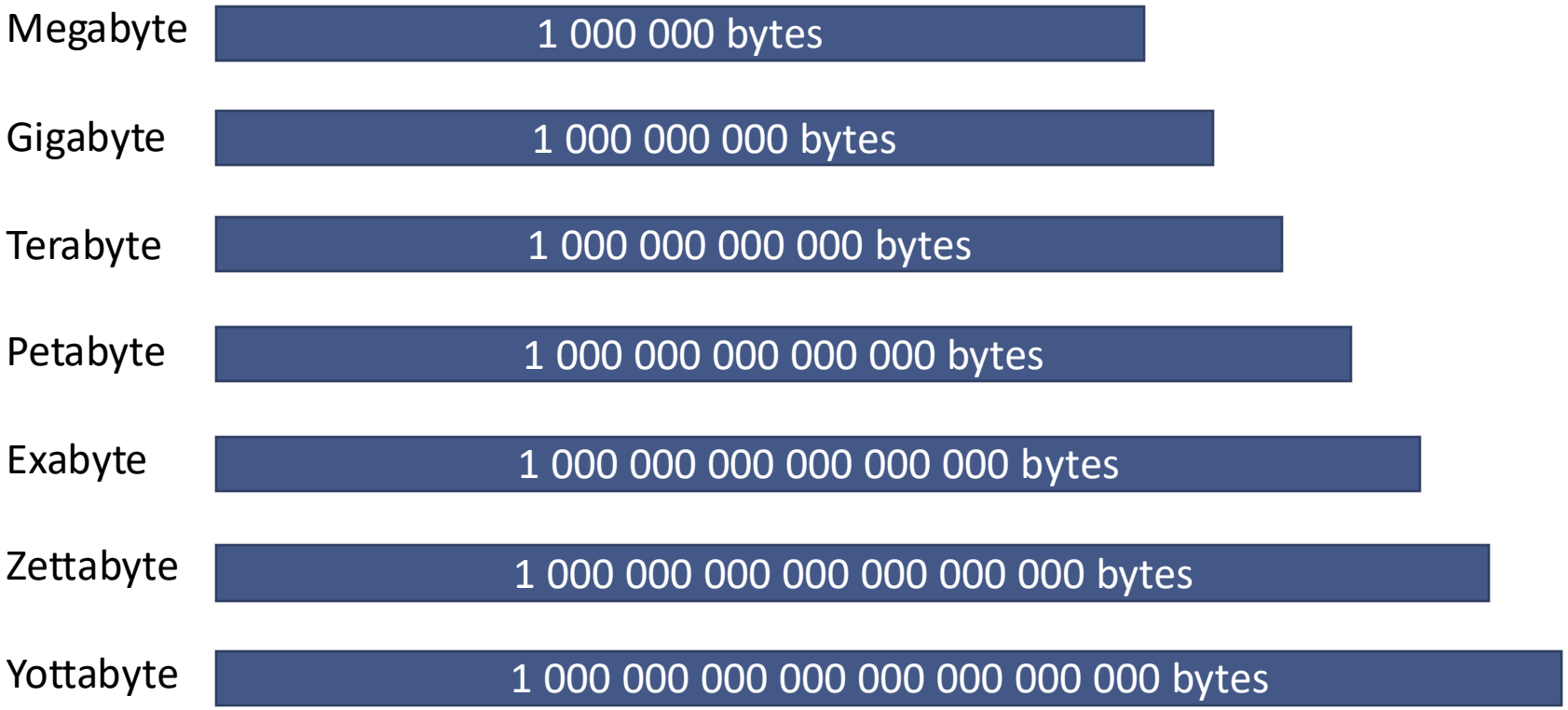
The Internet of Things, connected “smart” devices that interact with each other and us while collecting all kinds of data, is exploding (from **2 billion devices in 2006** to a projected **200 billion by 2020**) and is one of the primary drivers for our data vaults exploding as well.

Let’s take a look at just some of the stats and predictions for just one type of device, voice search:

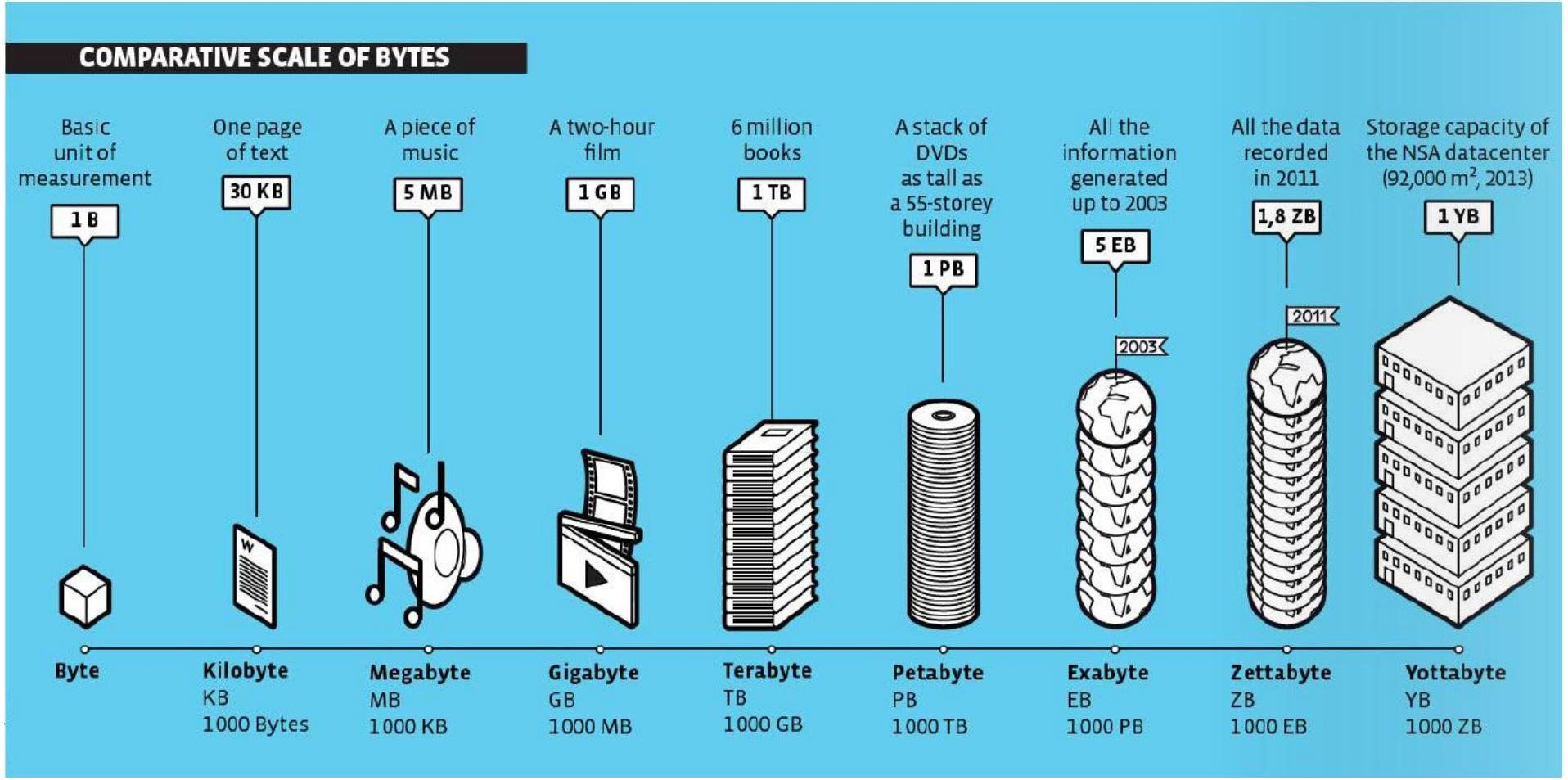
- There are 33 million voice-first devices in circulation
- 8 million people use voice control each month
- Voice search queries in Google for 2016 were up 35 times over 2008

https://www.forbes.com/sites/bernardmarr/2018/05/21/how-much-data-do-we-create-every-day-the-mind-blowing-stats-everyone-should-read/#559c6dee60ba

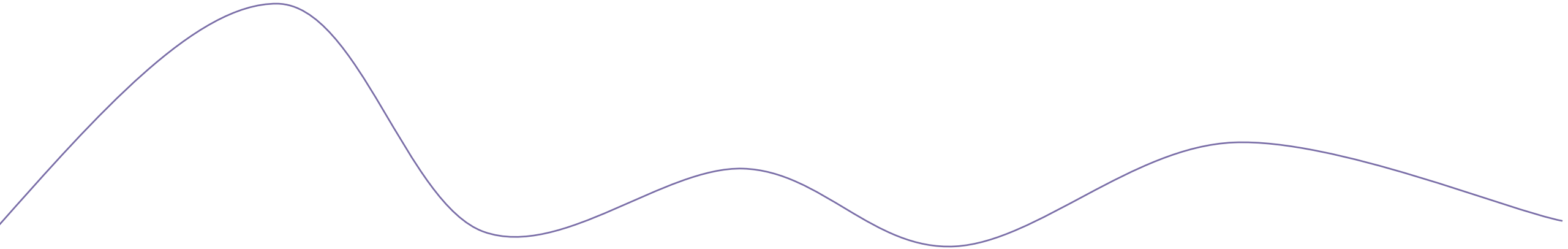
Scale it


Megabyte
1 000 000 bytes
Gigabyte
1 000 000 000 bytes
Terabyte
1 000 000 000 000 bytes
Petabyte
1 000 000 000 000 000 bytes
Exabyte
1 000 000 000 000 000 000 bytes
Zettabyte
1 000 000 000 000 000 000 000 bytes
Yottabyte
1 000 000 000 000 000 000 000 000 bytes

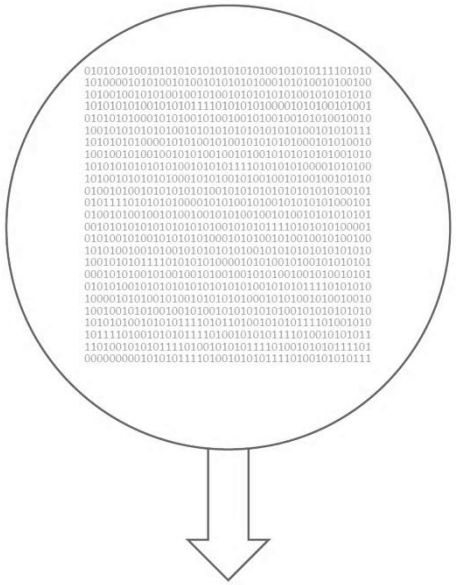
Scale it


COMPARATIVE SCALE OF BYTES
Basic unit of measurement
1 B
Byte
One page of text
30 KB
Kilobyte
KB
1000 Bytes
A piece of music
5 MB
Megabyte
MB
1000 KB
A two-hour film
1 GB
Gigabyte
GB
1000 MB
6 million books
1 TB
Terabyte
TB
1000 GB
A stack of DVDs as tall as a 55-storey building
1 PB
Petabyte
PB
1000 TB
All the information generated up to 2003
5 EB
2003
Exabyte
EB
1000 PB
All the data recorded in 2011
1,8 ZB
2011
Zettabyte
ZB
1000 EB
Storage capacity of the NSA datacenter (92,000 m², 2013)
1 YB
Yottabyte
YB
1000 ZB

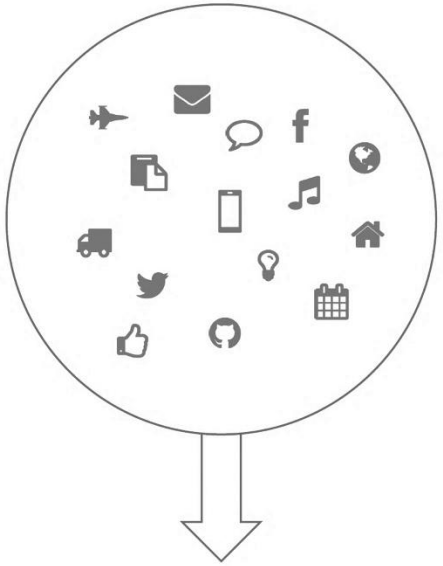
No SQL



Limit of relational world



More and more data



Variety of data

Limit of SQL

- Could RDBMSs be distributed, or were they capacity-limited to the hardware they resided on?
- How did distribution affect throughput?
- Could RDBMSs support multi-tenancy, and did this threaten transaction isolation?
- Could relational systems support very wide columns, variable-length records, semi- or even unstructured data?

The birth of NoSQL

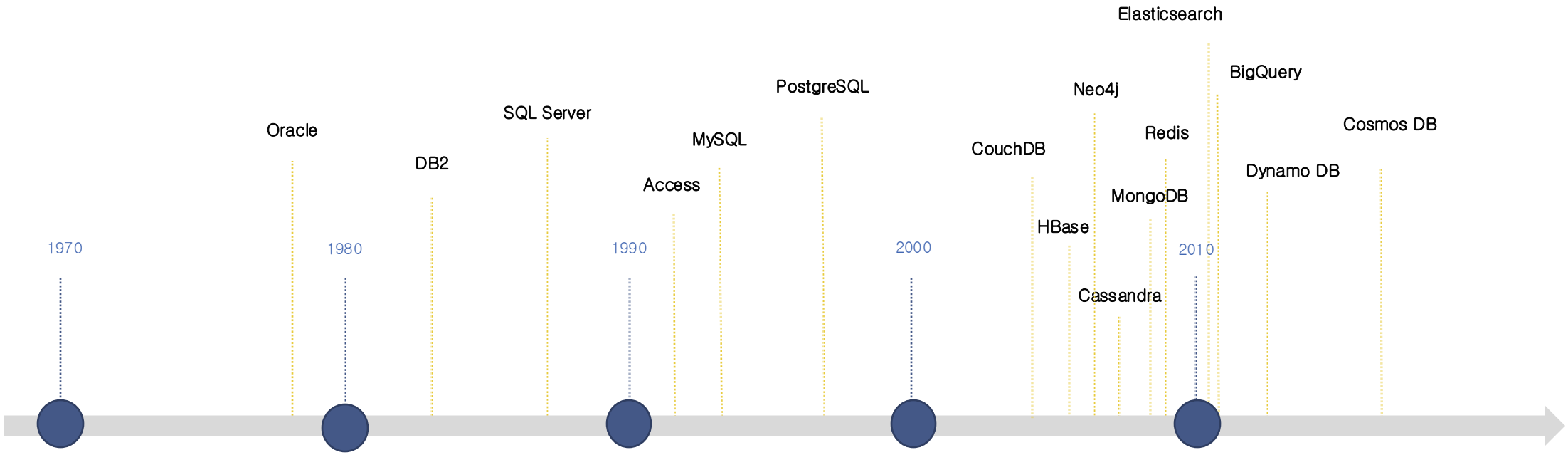
No SQL :

- Non relational databases
- Not Only SQL
- NoSQL

Why?

- Handle large volumes of rapidly changing structured, semi-structured and unstructured data.
- Suitable for agile sprints, fast iterations of schemas and frequent pushes (CI/CD)
- API & object programming oriented allowing for more flexible and easier use
- Distributed technology

Numerous databases ...



… and various models

Relational data store



HDFS



Key Value data store



Columnar data store



Object store



Search data store



Graph data store

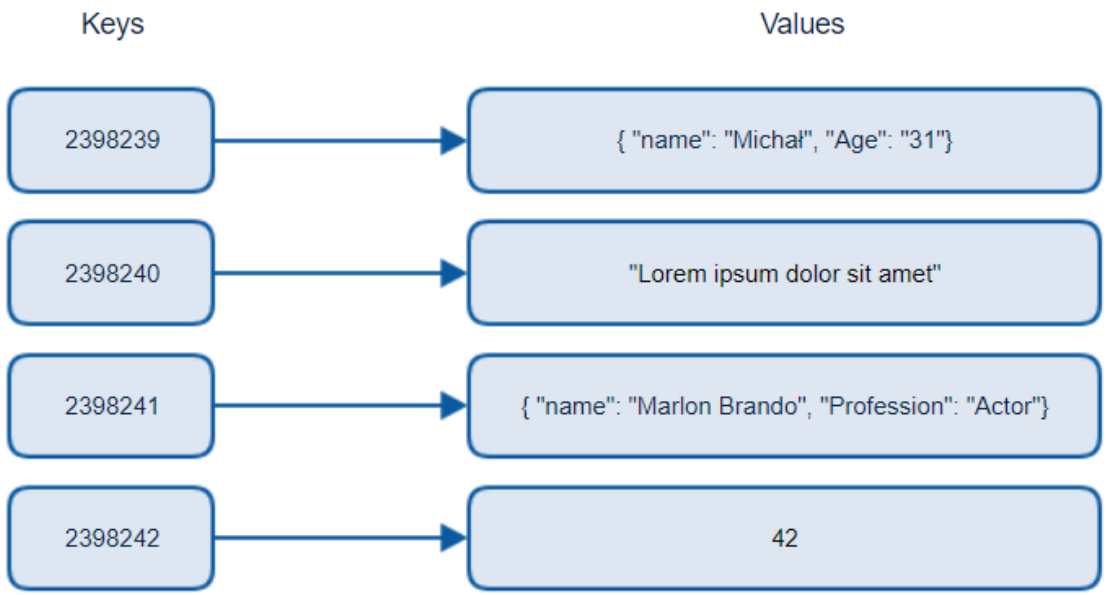


Document data store



Key value

Key-value databases are organized into partitions or buckets, which can contain one or several entities, and each record is represented by a unique row key. Records in turn contain one or multiple fields.



Key value

- Scalability
- Very fast read speed
- Simple and flexible data model

- No relationship between entities
- No transaction-like behaviour
- Only per-key queries are supported
- No support for retrieving multiple keys at once
- Slow multiple updates and collection scans

<u>Uses Cases:</u>

- Web session cache
- Store user preferences

Key value







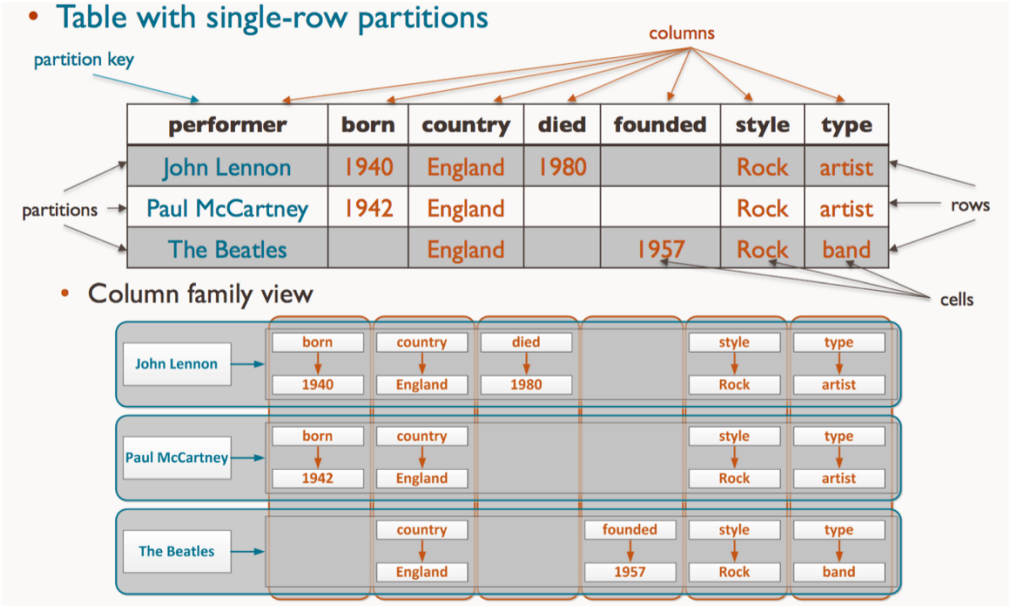






Columnar Data Store

Column family databases share concepts with both RDBMS and key-value stores. You can think of the rows as keys in a key-value store, and the columns as the value. Their optimal use case is for large data ingestion or data analytics, being suitable for storing billions of rows and tens of thousands of columns.



Columnar Data Store



- Scalability
- Fast write/read

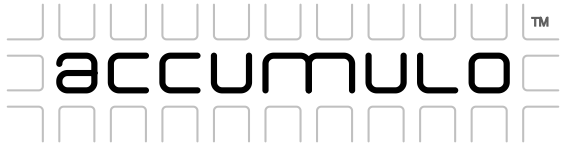


- Update/modification operations are slow

<u>Uses Cases:</u>

- Telemetry
- IoT
- Reporting

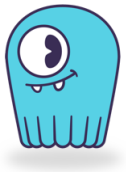
Columnar Data Store







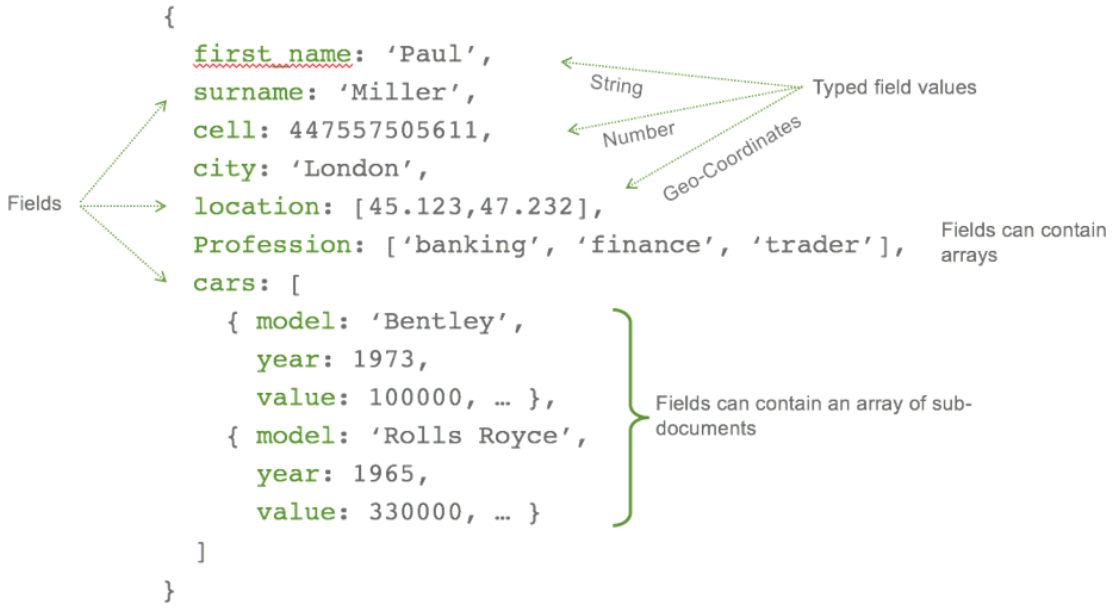






Document Databases

Document Store databases build on the concept behind key-value, extending it to support complex multi-layered objects named documents.



Document Databases

- Scalability for complex objects
- Document-oriented data model, JSON or XML allows for complex and schema-less structure
- Supports queries and joins within the document
- Data modeling paradigm allows storing all the data in a single document
- Fast read and writes

- Data modeling paradigm leads to having data duplicated among documents
- Complex design leads to inconsistency

<u>Uses Cases:</u>

- Social networks
- eCommerce
- Anything in which you can relax ACID compliance

Document Databases

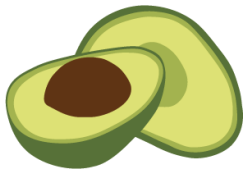












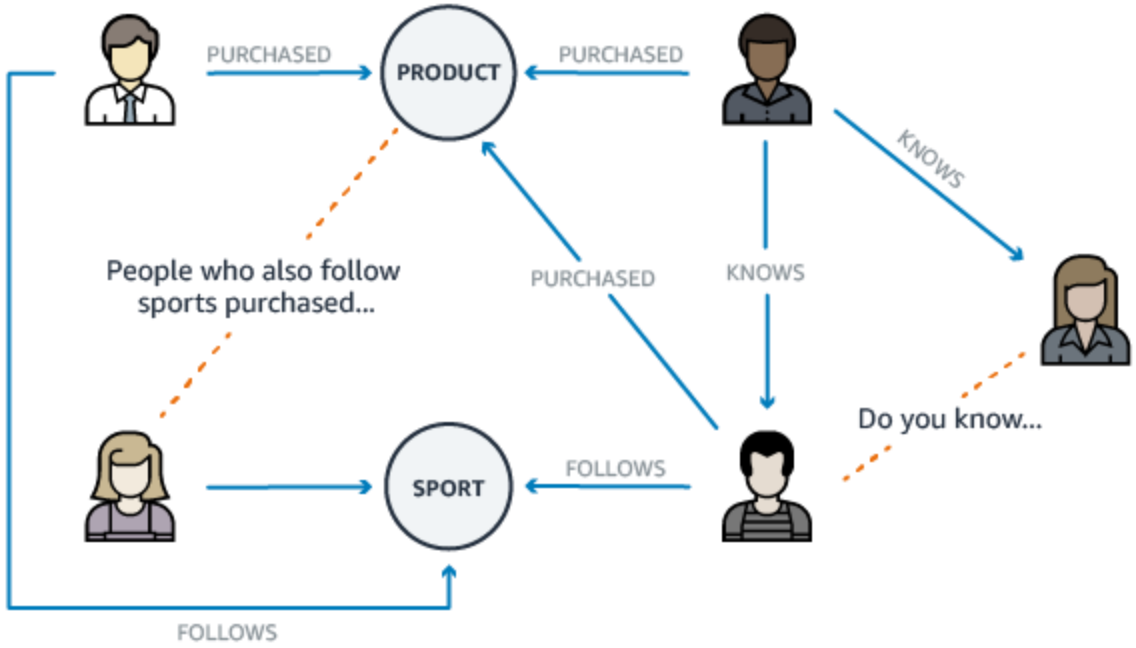




Graph-Like Data Models

Graph database types shift the focus onto the relationship between entities. Entities, such as users, are represented by nodes, whereas the connections between entities dictate how they are related.

Graph databases store the relationship information within each node, by doing so, the need for lookup operations needed in relational databases is removed, saving much-needed resources.



Graph-Like Data Models

- The structures are agile and flexible.
- The representation of relationships between entities is explicit.
- Queries output real-time results. The speed depends on the number of relationships.

- There is no standardized query language. The language depends on the platform used.
- Graphs are inappropriate for transactional-based systems.
- The user-base is small, making it hard to find support when running into a problem

Uses Cases:

- Knowledge Graphs
- Identity Graphs
- Fraud Detection
- Recommendation Engines
- Social Networks

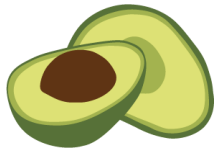
Graph-Like Data Models













OpenClassroom

Maîtrisez les bases de données NoSQL

15 heures Moyenne

https://openclassrooms.com/fr/courses/4462426-maitrisez-les-bases-de-donnees-nosql

Table des matières

Partie 1 - Immergez vos données dans le NoSQL

1. Choisissez votre famille NoSQL
2. Maitrisez le théorème de CAP
3. Passez à l'échelle
4. Choisissez votre base de données NoSQL

Quiz : Savez-vous vraiment ce qu'est le NoSQL ?

Partie 2 - Administrez vos données avec MongoDB

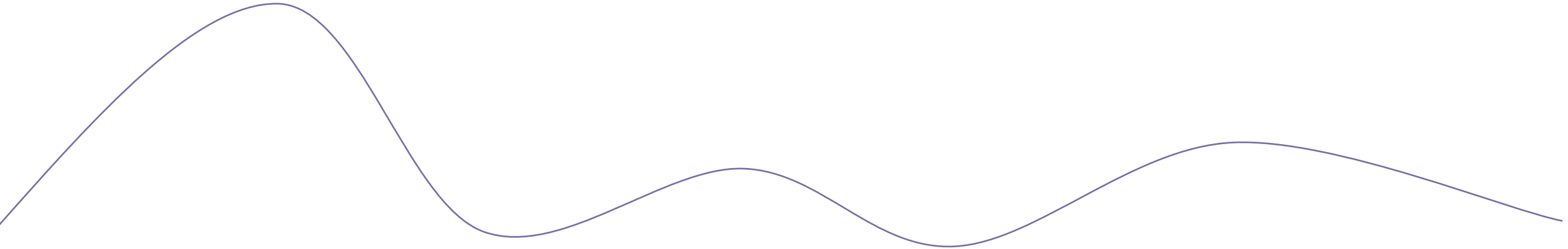
1. Découvrez le fonctionnement de MongoDB
2. Interrogez vos données avec MongoDB
3. Protégez-vous des pannes avec les ReplicaSet
4. Distribuez vos données avec MongoDB
5. Entraînez-vous à créer et à interroger une base de données MongoDB

Partie 3 - Visualisez vos données avec ElasticSearch et Kibana

1. Etudiez le fonctionnement d'Elasticsearch
2. Interrogez des données textuelles
3. Faites grandir votre base
4. Visualisez et prototypez avec Kibana
5. Entraînez-vous à extraire l'essence d'une base de données

Certificat de réussite (voir un exemple)

The Hadoop Ecosystem



WHAT IS BIG DATA?

"Big Data are a collection of data sets so large and complex that it becomes difficult to process using on-hand database management tools or traditional data processing applications."

Tom White – Hadoop Definitive Guide.

Big Data

The word « Big Data » is very recent : 2008.

More than technologies, it defined a new era in IT, called first Petabyte Age.

The five V's of big data

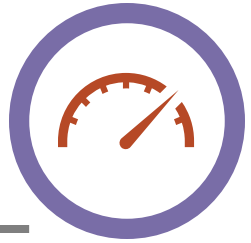


Value

The ability to transform a tsunami of data into business.

Veracity

Equivalent to quality. Are the data "clean" and accurate?



Velocity

Sometimes it's better to have limited data in real time than lots of data at a low speed.



Volume

Green marketing is a practice whereby companies seek to go.



Variety

A company can obtain data from many different sources: from in-house devices to smartphone GPS technology or what people are saying on social networks.

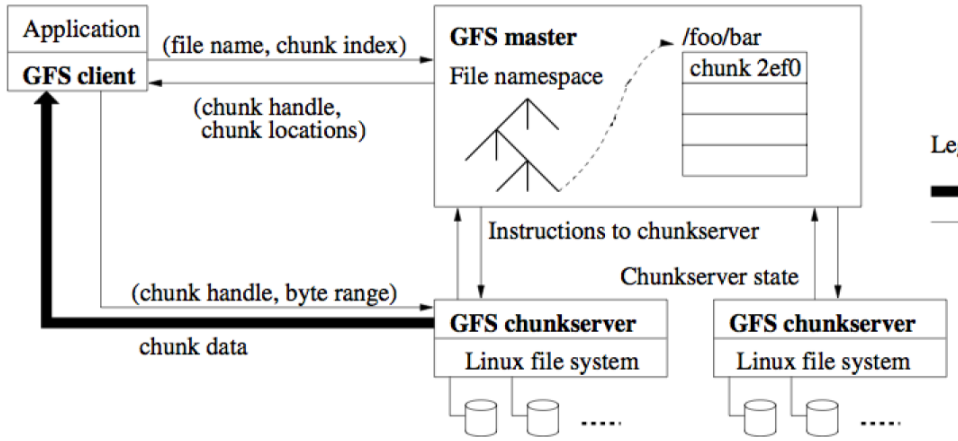
Hadoop



Created by Doug Cutting

2003

Google publish the GFS architecture



2004

- NDFS implementation (open source version of GFS)
- Google publishes MapReduce algorithm

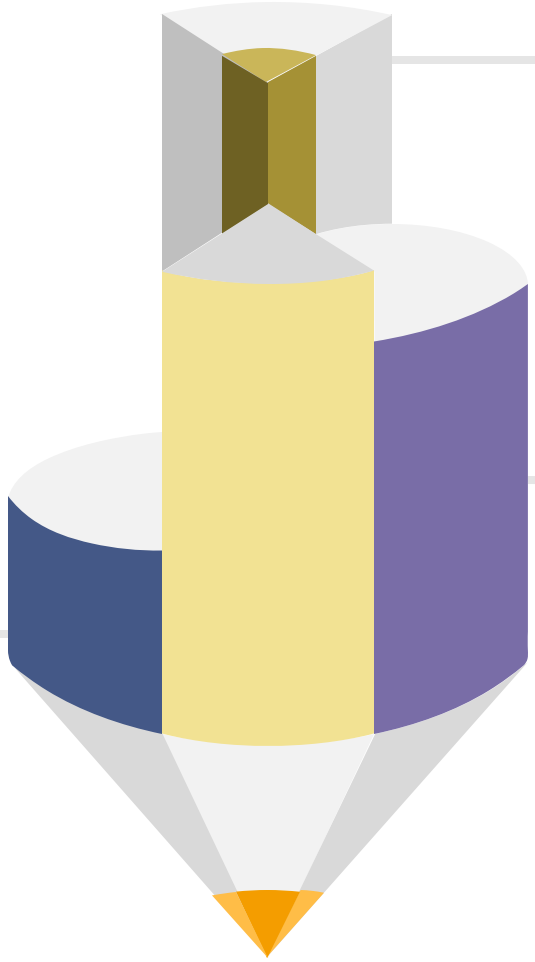
2008

Yahoo! uses Hadoop in production

ıe Hadoop

- Doug Cutting join Yahoo!

Hadoop



May 2009

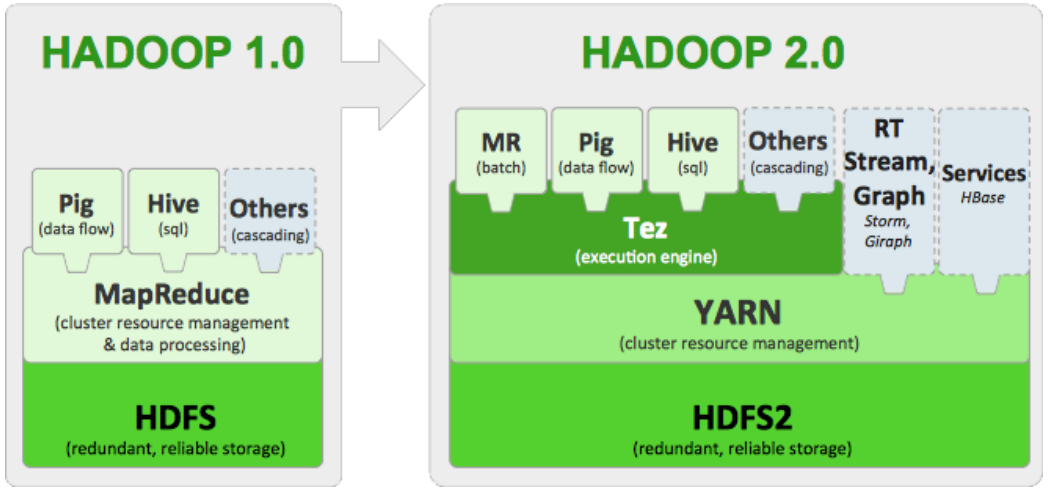
Yahoo! 62 secondes

November 2008

Google optimizes the treatment for reach 68 seconds

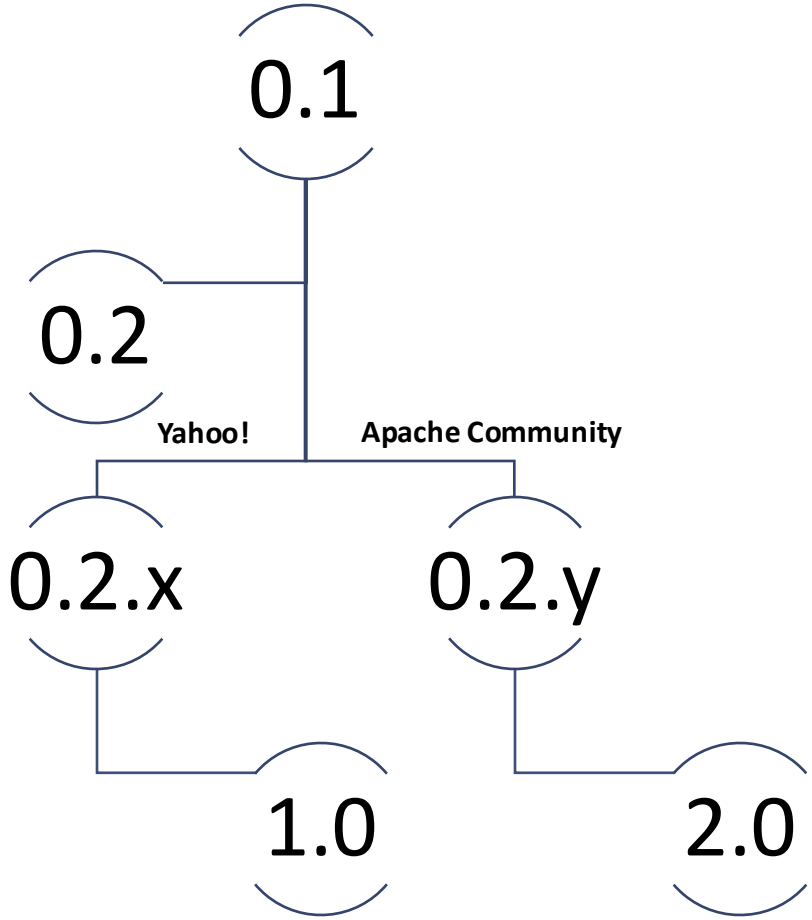
April 2008

Hadoop's ecosystem



Attributes	Hadoop 2.x	Hadoop 3.x
Handling Fault-tolerance	Through replication	Through erasure coding
Storage	Consumes 200% in HDFS	Consumes just 50%
Scalability	Limited	Improved
File System	DFS, FTP and Amazon S3	All features plus Microsoft Azure Data Lake File System
Manual Intervention	Not needed	Not needed
Scalability	Up to 10,000 nodes in a cluster	Over 10,000 nodes in a cluster
Cluster Resource Management	Handled by YARN	Handled by YARN
Data Balancing	Uses HDFS balancer for this purpose	Uses Intra-data node balancer

Hadoop's ecosystem



Hadoop's distribution



January 2019

cloudera

HORTONWORKS®

MAPR®

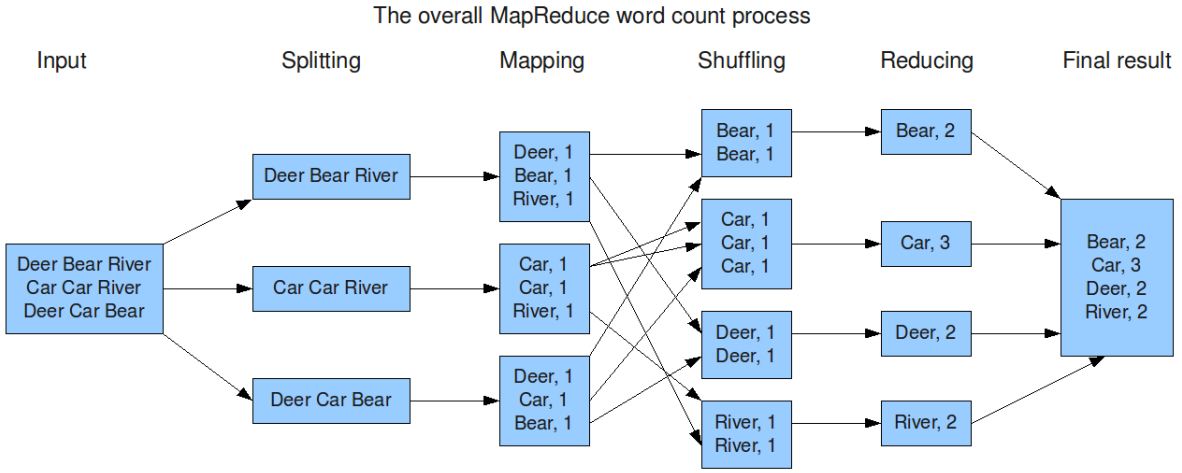
MapReduce

MapReduce is a programming paradigm that enables massive scalability across hundreds or thousands of servers in a Hadoop cluster. The MapReduce concept is simple to understand for those who are familiar with clustered scale-out data processing solutions.

MapReduce split the work in 2 phases: Map & Reduce. Each phase has inputs and outputs in key-value format.

MapReduce

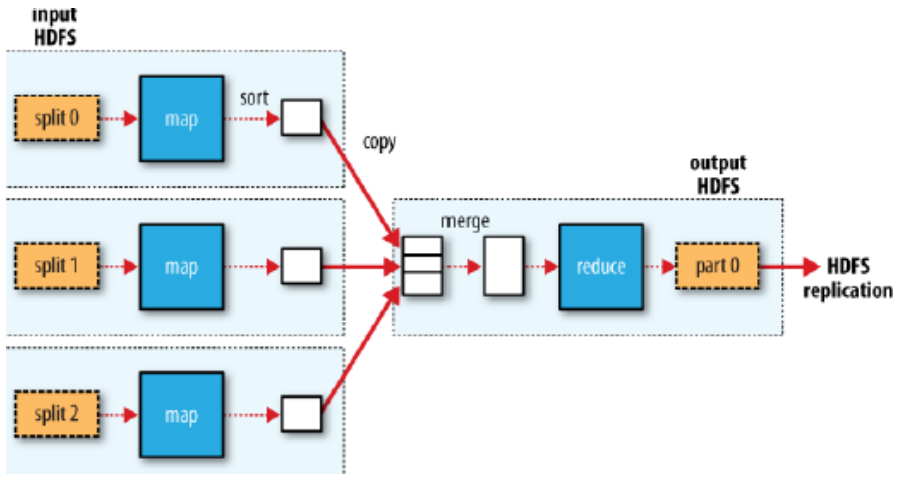
- Splitting: First, we separate all data in different threads. Each thread can calculate more quickly than one more powerful processor.
- Map: It affects one key for one value to have the couple <key, value>.
- Shuffle: All data are group by key.
- Reduce: Involve all the values corresponding to the same key to a unique pair.



MapReduce

Hadoop slip his task in map and reduce tasks. Two kind of node control the execution:

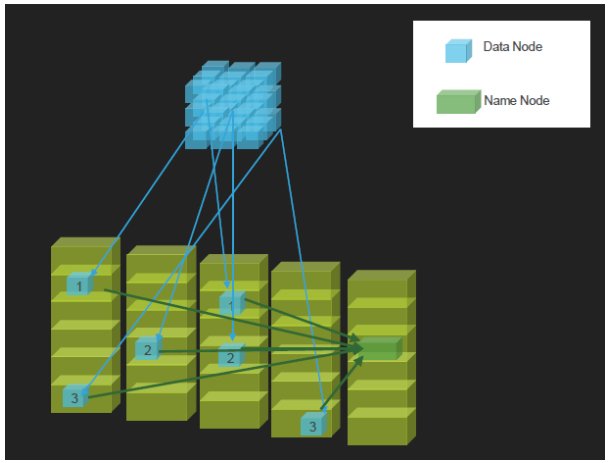
- Jobtracker : allocation of resources, distribution of treatment
- Tasktrackers : supervision of the execution and returns monitoring data



HDFS

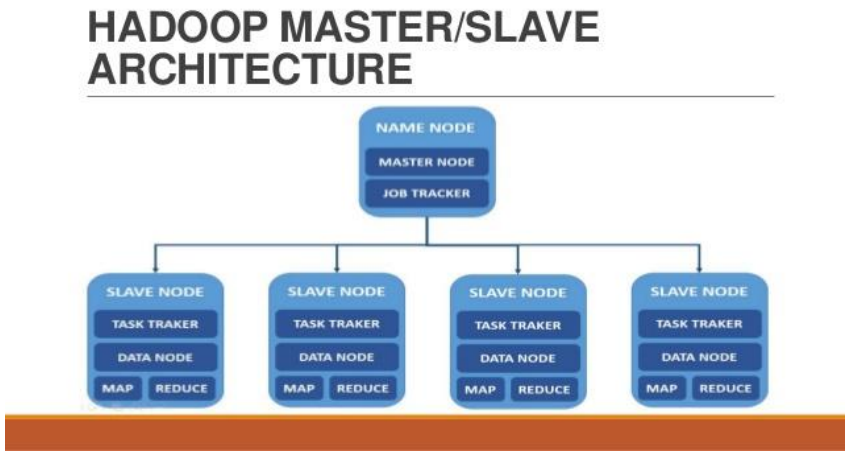
HDFS : Hadoop Distributed File System.

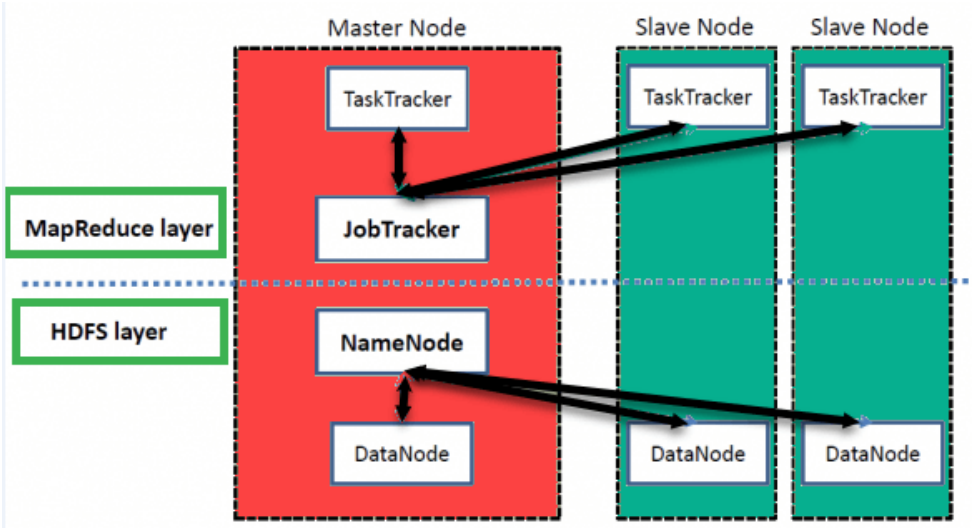
- 64Mo default bloc
- 3 replica at least



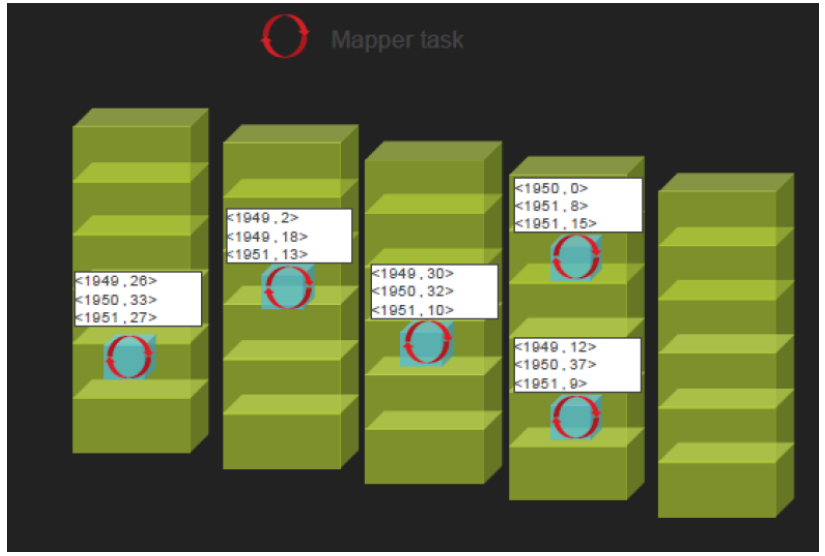


HDFS


HADOOP MASTER/SLAVE ARCHITECTURE
NAME NODE
MASTER NODE
JOB TRACKER
SLAVE NODE
TASK TRAKER
DATA NODE
MAP
REDUCE
SLAVE NODE
TASK TRAKER
DATA NODE
MAP
REDUCE
SLAVE NODE
TASK TRAKER
DATA NODE
MAP
REDUCE
SLAVE NODE
TASK TRAKER
DATA NODE
MAP
REDUCE

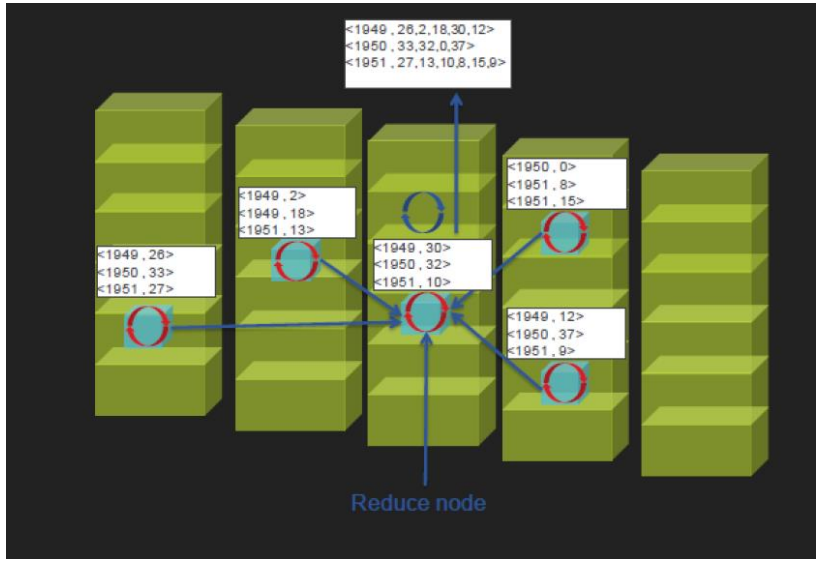

Master Node
Slave Node
Slave Node
TaskTracker
TaskTracker
TaskTracker
MapReduce layer
JobTracker
HDFS layer
NameNode
DataNode
DataNode
DataNode

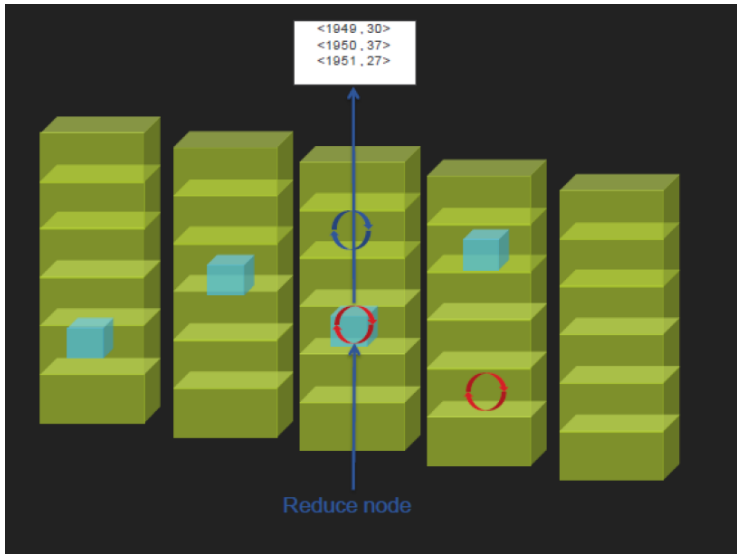
MapReduce



map

shuffle



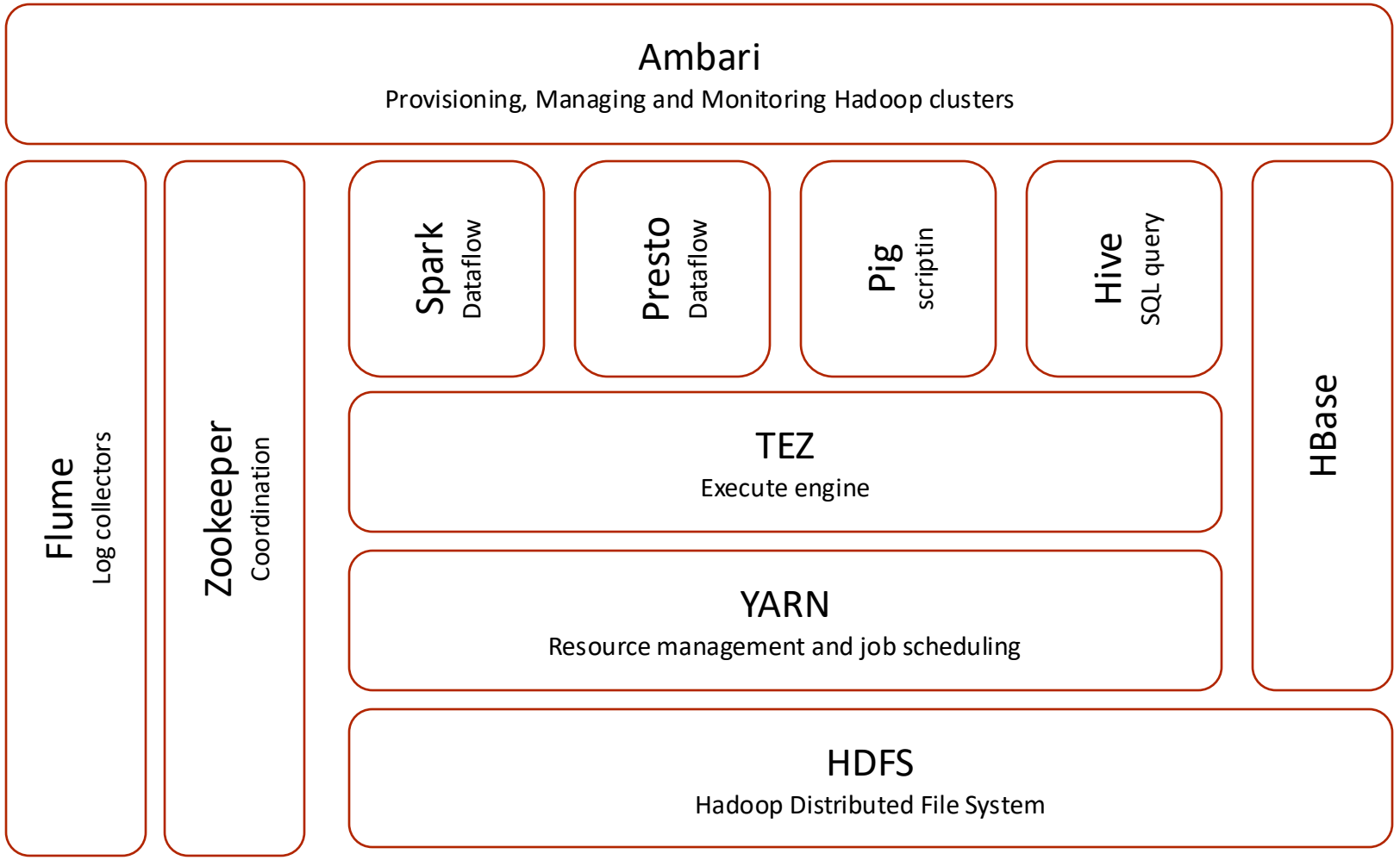


reduce

HDFS

Advantages	Disadvantages
○ Can store very large files ○ Features optimized algorithms to access data directly ○ Works on inexpensive hardware ○ Easily scalable	○ Not optimized for small files. ○ Not intended for the selection of a specific data ○ No update of the files (obligation to delete and rewrite the data) ○ No multiple entries

Hadoop Ecosystem



Hive



Hive is an ETL and Data warehousing tool developed on top of Hadoop Distributed File System (HDFS). Hive makes job easy for performing operations like:

- Data encapsulation
- Ad-hoc queries
- Analysis of huge datasets

Schema on read (Hive) :

- o Very fast initial loading (+).
- o Very flexible method (+).
- o No updates, no indexing (-).

Schema on write (SGBD-R) :

- o Good performance in querying (+).
- o Loading data long (-).

Pig



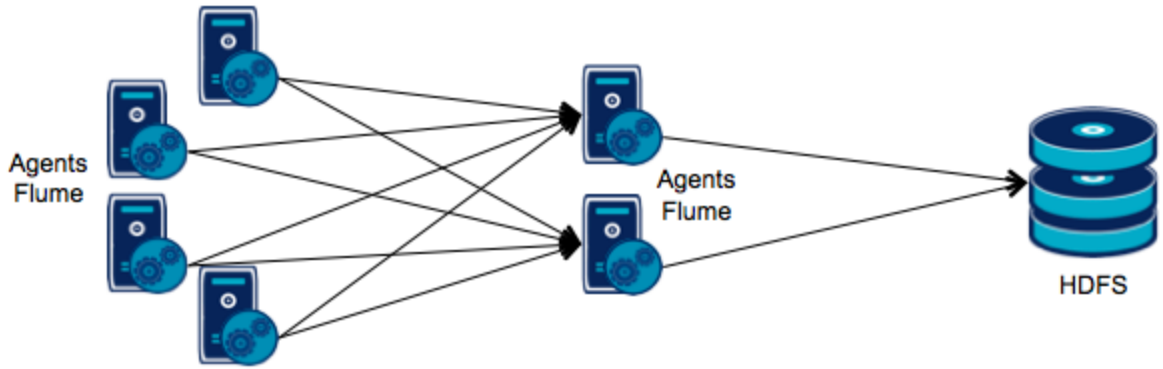
Pig is a high level scripting language that is used with Apache Hadoop. Pig enables data workers to write complex data transformations without knowing Java. Pig's simple SQL-like scripting language is called Pig Latin, and appeals to developers already familiar with scripting languages and SQL.

Pig is complete, so you can do all required data manipulations in Apache Hadoop with Pig. Through the User Defined Functions(UDF) facility in Pig, Pig can invoke code in many languages like JRuby, Jython and Java.

Flume

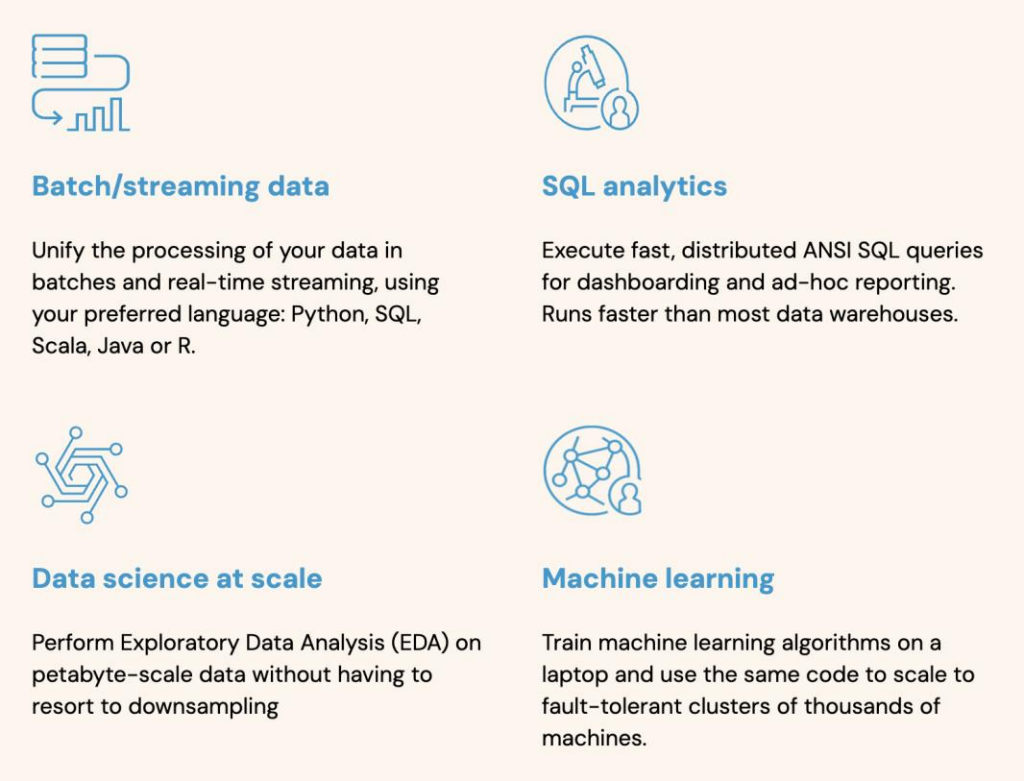


Flume is a scalable, real-time ingest framework that allows you to route, filter, aggregate, and perform "mini-operations" on data on its way into a scalable processing platform like CDH.



Spark

Apache Spark™ is a multi-language engine for executing data engineering, data science, and machine learning on single-node machines or clusters.

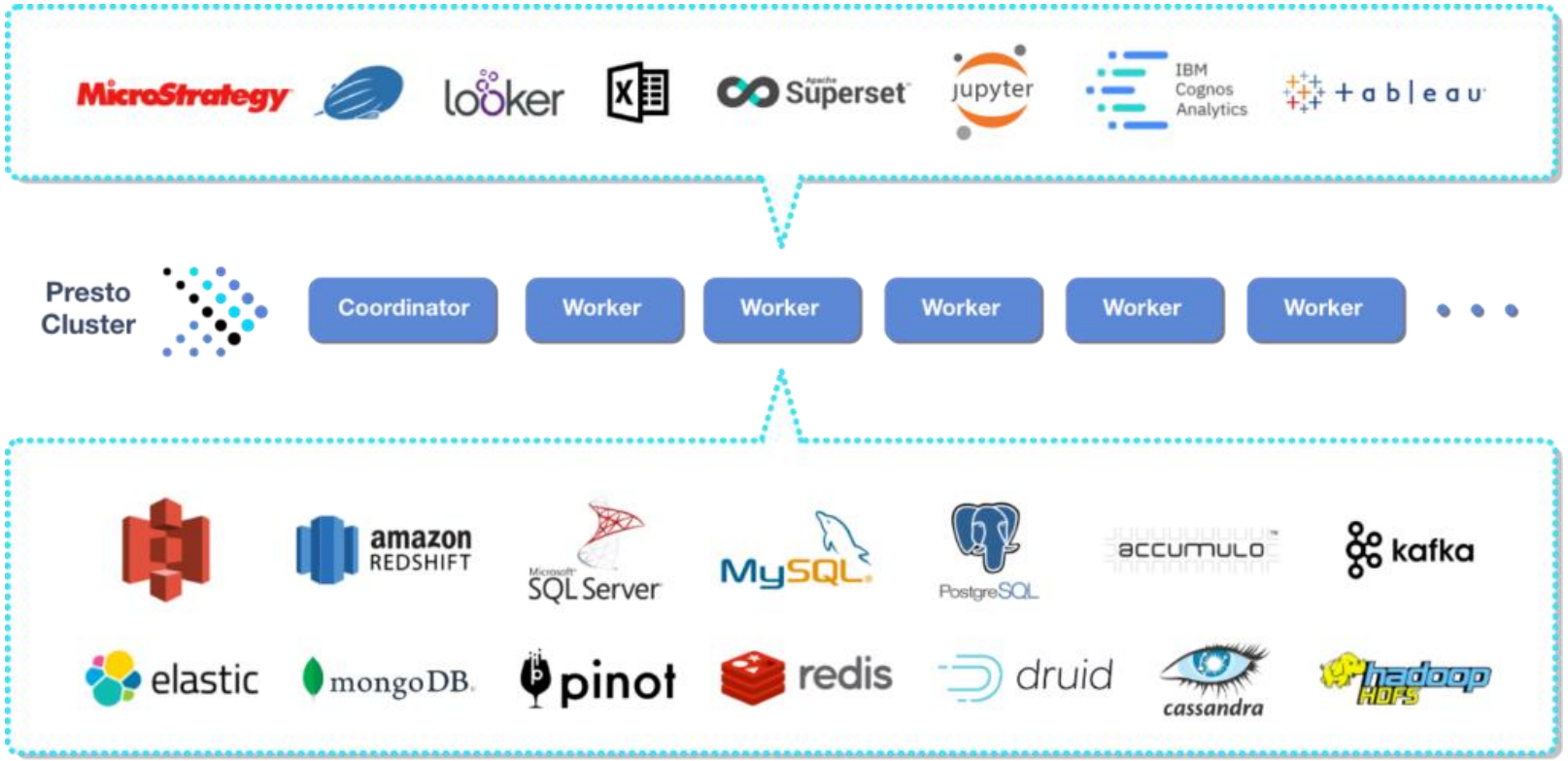


https://spark.apache.org/

Presto

presto

Presto is an open source SQL query engine that's fast, reliable, and efficient at scale. Use Presto to run interactive/ad hoc queries at sub-second performance for your high volume apps.



https://prestodb.io/

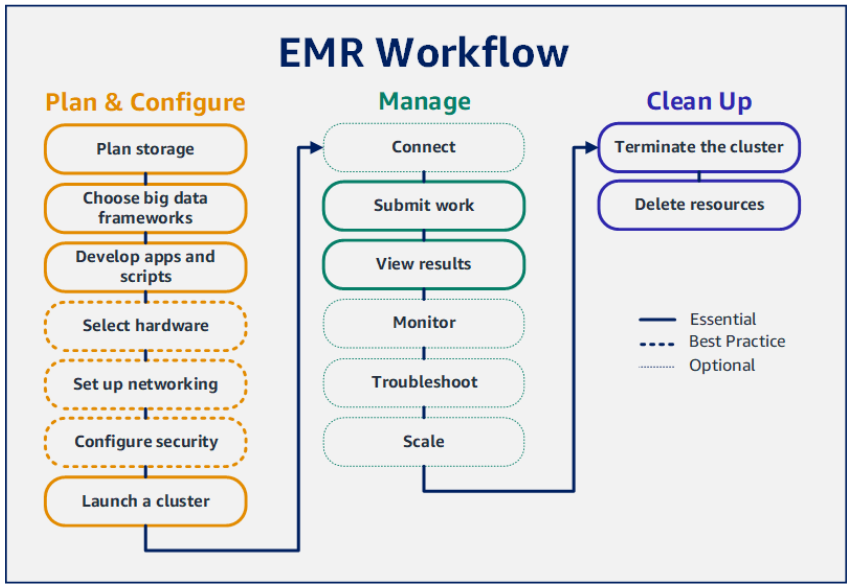
Reprise à 11h15



datalex

LAB – Getting started with Amazon EMR





1. Follow this first tutorial in order to start understanding EMR.

2. Then, walk through a workflow to quickly set up an Amazon EMR cluster and run a Spark application following this other tutorial.

AWS account



Your turn !

AWS Free Tier

Gain free, hands-on experience with the AWS platform, products, and services

Create a Free Account

Types of offers

Explore more than 60 products and start building on AWS using the free tier. Three different types of free offers are available depending on the product used. See below for details on each product.

Always free

These free tier offers do not expire and are available to all AWS customers

12 months free

Enjoy these offers for 12-months following your initial sign-up date to AWS

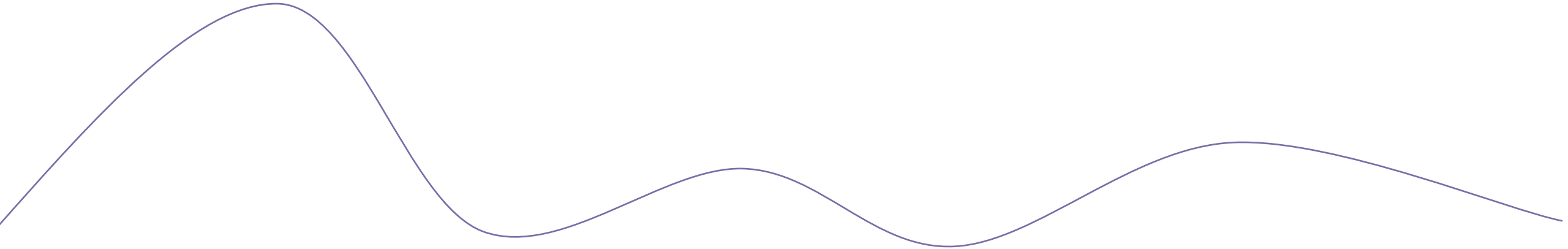
Trials

Short-term free trial offers are available through many different software solutions

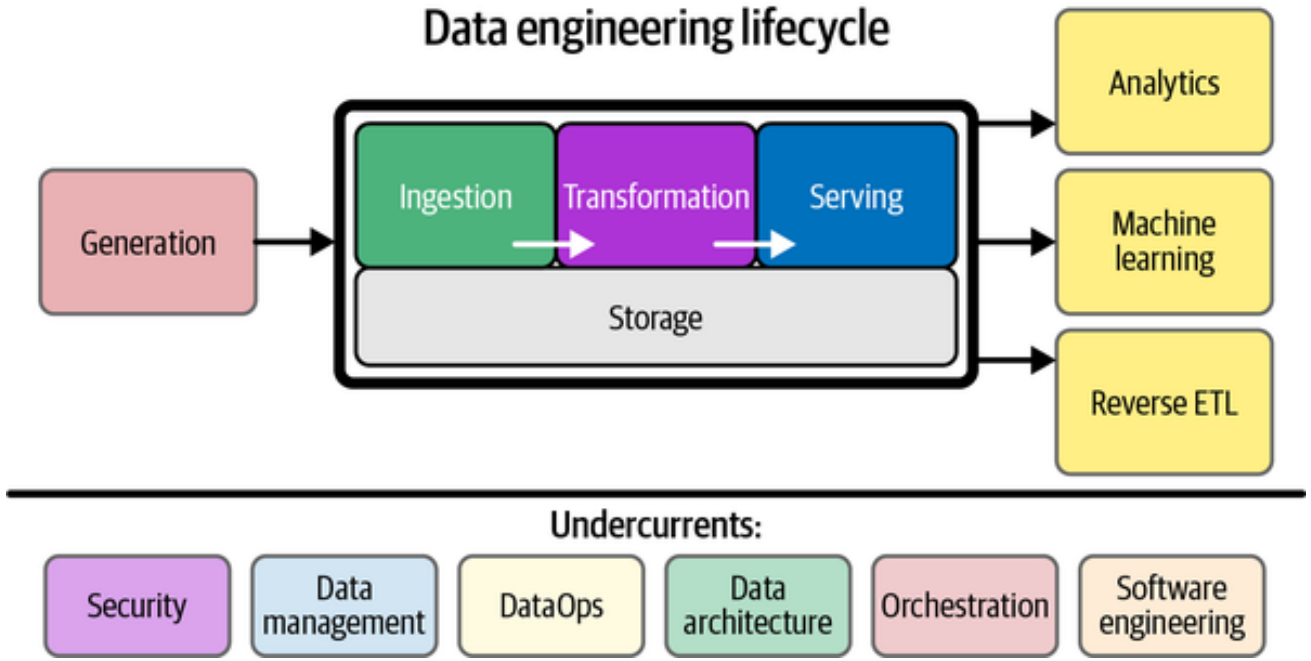
https://aws.amazon.com/free



The Data Engineering Lifecycle

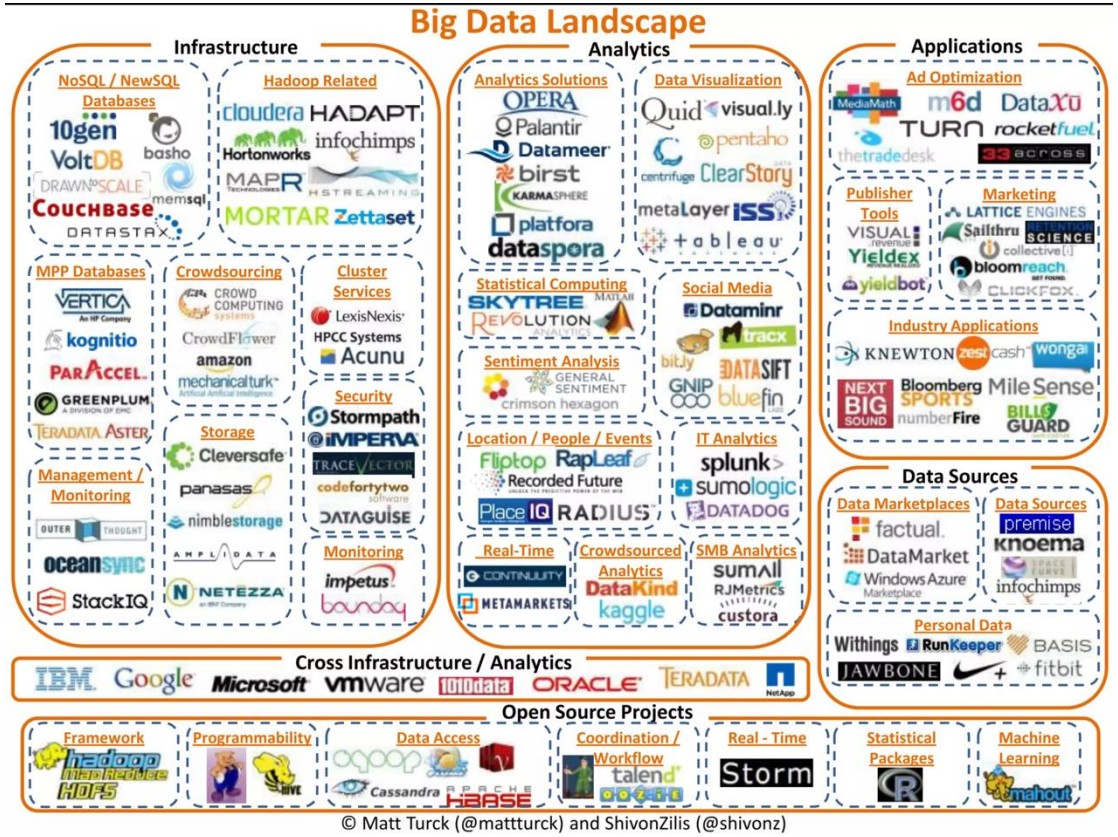


The Data Engineering Lifecycle



"fundamental of data engineering" by Joe Reis & Matt Housley

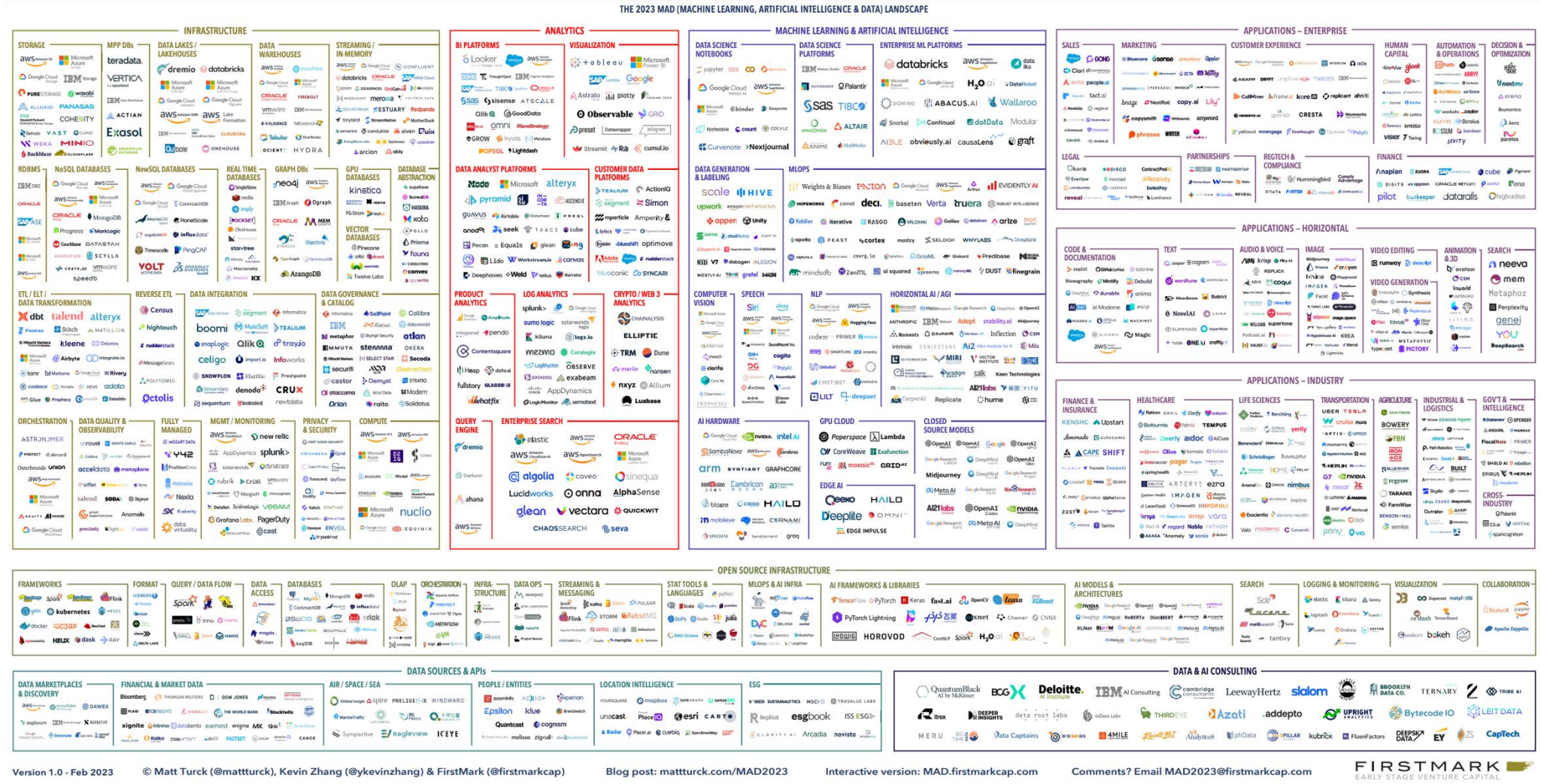
Data Landscape



© Matt Turck (@mattturck) and ShivonZilis (@shivonz)

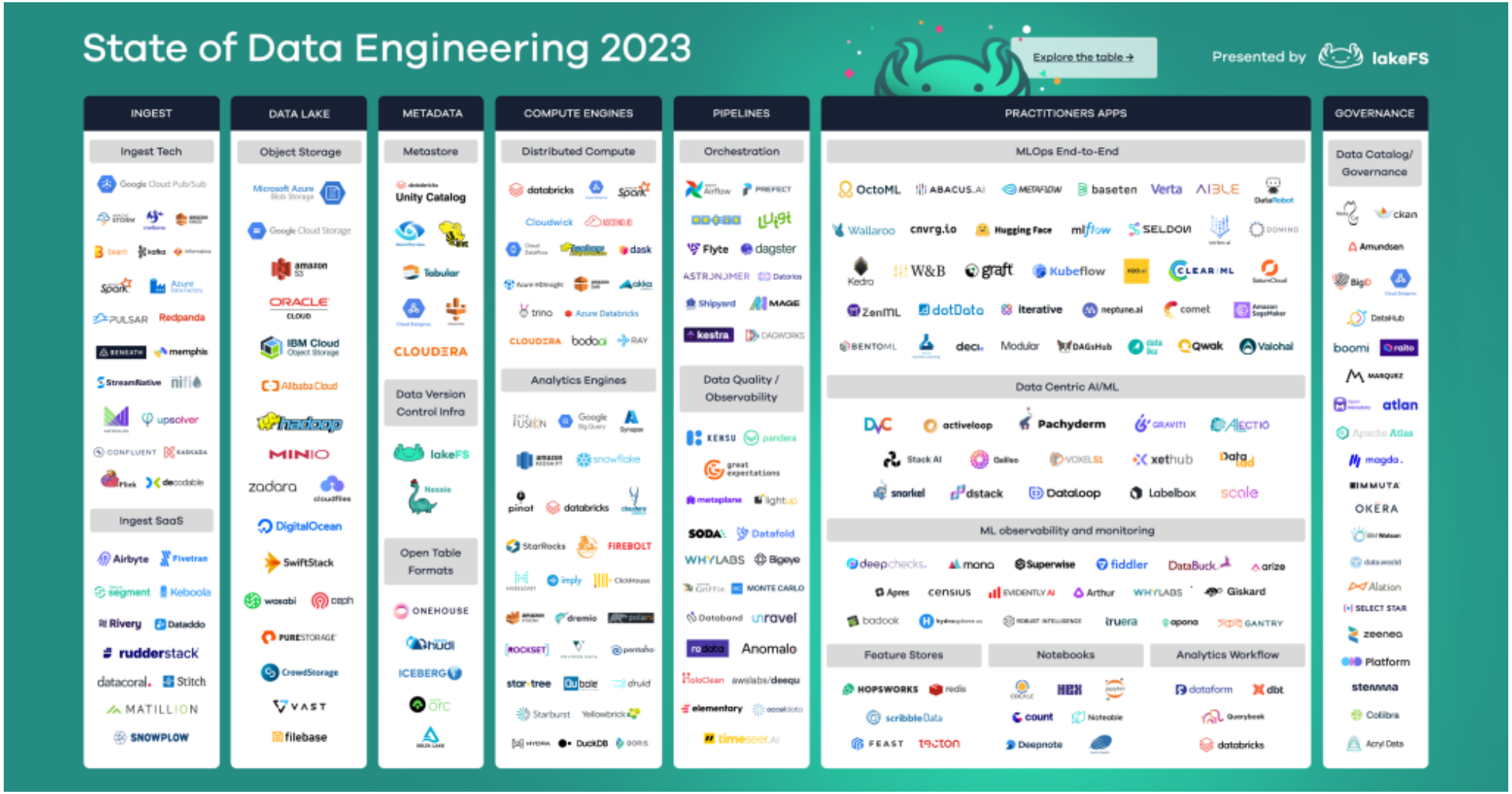
https://mattturck.com/a-chart-of-the-big-data-ecosystem/

Data Landscape



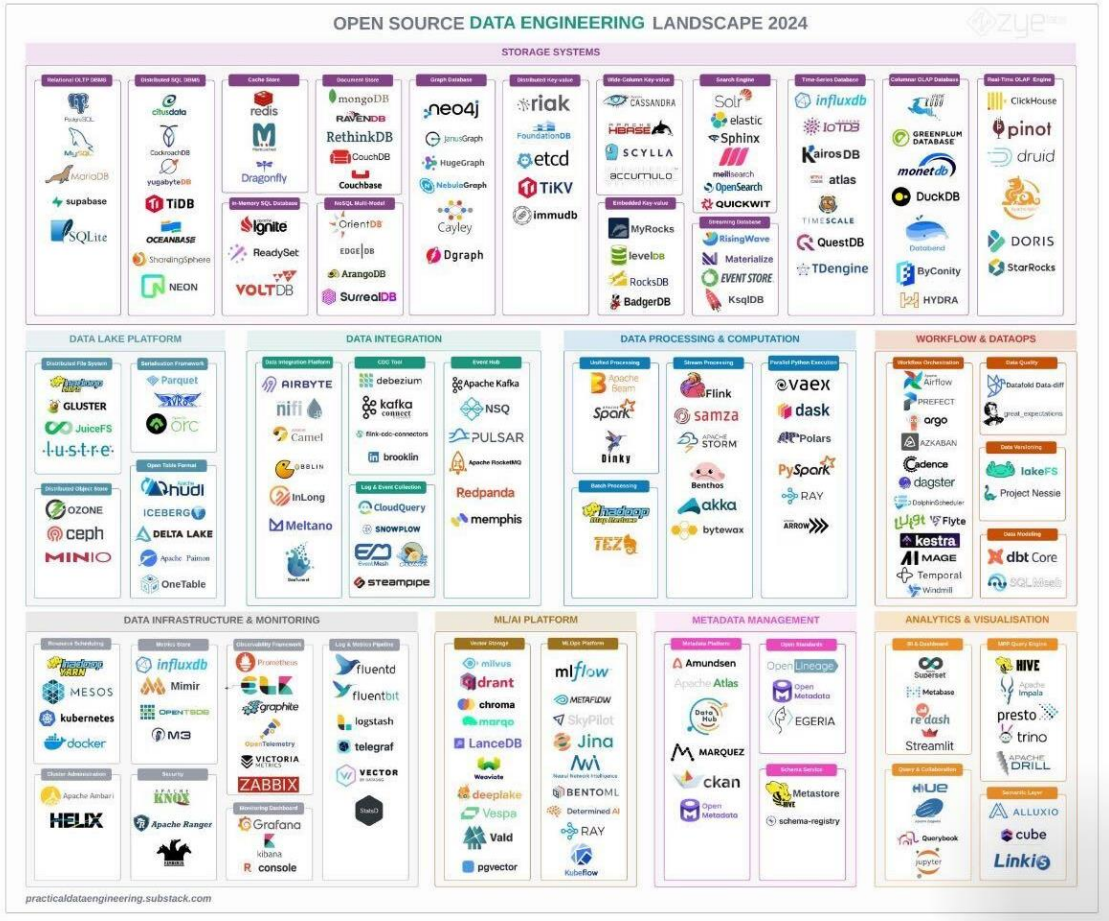
https://mattturck.com/MAD2023/

State of Data Engineering



https://lakefs.io/blog/the-state-of-data-engineering-2023/

State of Data Engineering



https://practicaldataengineering.substack.com/p/open-source-data-engineering-landscape

Main data jobs

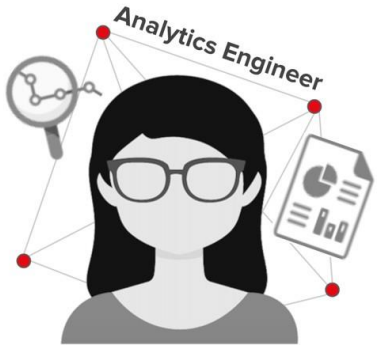
Data engineers primarily provision data stores. They make sure that massive amounts of data are securely and cost-effectively extracted, loaded, and transformed.

Data analyst enables businesses to maximize the value of their data assets through visualization and reporting tools. Their responsibilities also include designing and building scalable and effective data models, and enabling and implementing the advanced analytics capabilities into reports for analysis.

Data scientists / AI engineer perform advanced analytics to extract value from data. Their work can vary from descriptive analytics to predictive analytics.

Database administrator implements and manages the operational aspects of cloud-native, on-premise and hybrid data platform solutions.

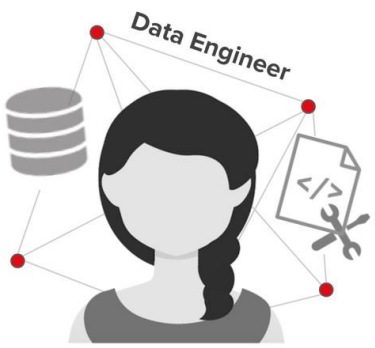
… get more specifics



tools: Sublime, Atom, Tableau
languages: SparkSQL, Presto, Python



tools: Jupyter, RStudio, PyCharm
languages: Python, Presto, R, PySpark



tools: IntelliJ, PyCharm, Sublime
languages: Scala, Spark, Python, SQL

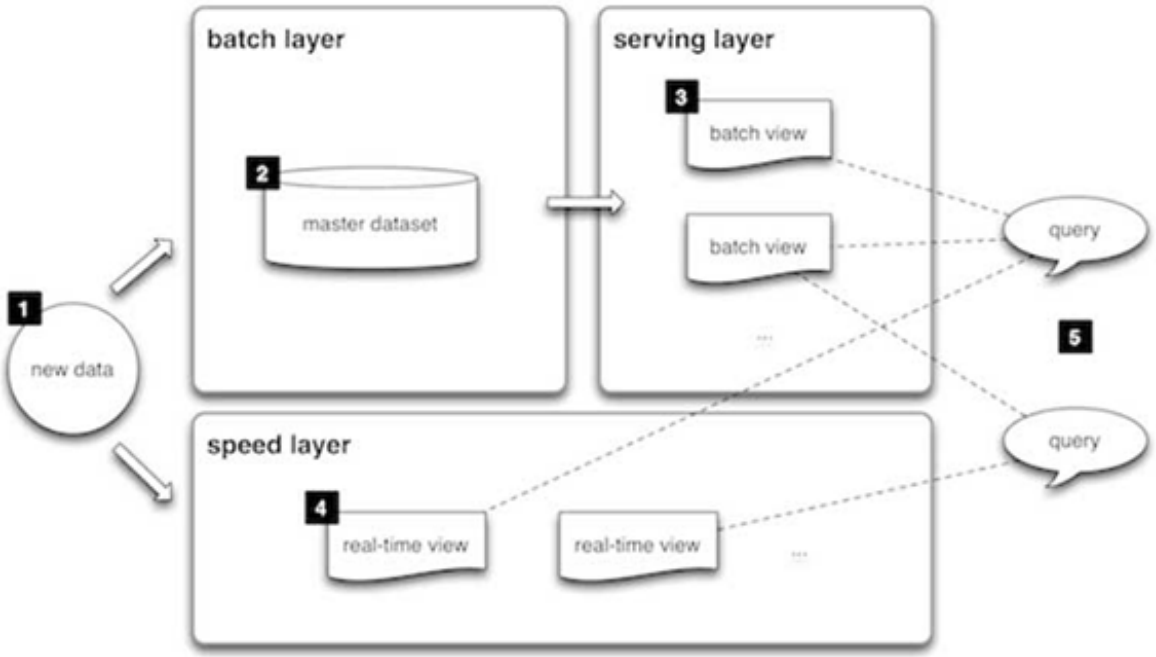
Data roles at Netflix

- business analyst
- data analyst
- quantitative analyst
- algorithm engineer
- analytics engineer
- data engineer
- data scientist
- machine learning scientist
- research scientist

https://medium.com/netflix-techblog/notebook-innovation-591ee3221233

Lambda architecture

Lambda Architecture is a data processing architecture designed to handle massive quantities of data by combining both batch and real-time processing methods. It aims to provide a comprehensive framework for processing and analyzing large data sets in a fault-tolerant and scalable manner.



Lambda architecture

The architecture typically consists of three main layers:

1. **Batch Layer**: This layer processes large volumes of data in batch mode, typically using a distributed processing system like Hadoop. It handles historical data, performs computations, and generates batch views that provide comprehensive analytics.

2. **Speed Layer**: Also known as the real-time layer, this component processes data in real-time or near-real-time, focusing on low-latency analytics. It uses technologies like Apache Storm or Apache Spark Streaming to provide immediate insights from incoming data streams.

3. **Serving Layer**: This layer merges the outputs from both the batch and speed layers, providing a unified view of the data for querying and analysis. It typically serves as an interface for users to access the data and insights generated by the other layers.

By combining batch and real-time processing, Lambda Architecture allows organizations to achieve both comprehensive historical analysis and immediate insights, making it suitable for applications that require timely decision-making based on large data sets.

ETL / ELT

ETL (Extract, Transform, Load) and ELT (Extract, Load, Transform) are data integration processes used to consolidate data from multiple sources into a target system, typically a data warehouse.

- ETL: In this process, data is first extracted from various sources, then transformed into a suitable format or structure, and finally loaded into the target system. This method is ideal for environments where data needs significant processing before being stored.

- ELT: In this approach, data is extracted from the source and loaded directly into the target system first, where it is then transformed. This method is more common in cloud-based data warehousing solutions, as it allows for faster loading and leverages the target system's processing power for transformations.

In summary, ETL transforms data before loading it, while ELT loads raw data and transforms it afterward.

ETL process

A typical ETL process includes the following steps:

1. **Extraction:** You can connect to any data store:
   - Traditional databases, like PostgreSQL, SQL Server, and MySQL.
   - Message brokers, like Kafka and Kinesis.
   - Distributed databases, like Cassandra and Redshift.
   - Data warehouses, like Hive and Azure Cosmos DB.
   - File types, like CSV, Parquet, and Avro.

2. **Data validation:** One aspect of ETL jobs is to validate that the data is what you expect. The data-validation step primarily includes determining:
   - Approximately the expected number of records.
   - Whether the expected fields are present.
   - That there are no unexpected missing values.

3. **Transformation:** This step generally includes applying structure and a schema to your data so that you can transform it into the desired format. Schemas can be applied to tabular data, such as data found in CSV files or relational databases, or to semistructured data, such as JSON data.

ETL process

4. **Corrupt record handling:** Handling bad data is also an important part of the entire ETL process. The built-in functions allow you to handle corrupt data, such as:
   - Missing and incomplete information.
   - Schema mismatches.
   - Differing formats or data types.
   - User errors when creating data producers.

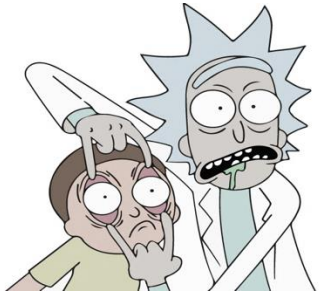
5. **Loading data:** Once transform you can save the resultat to different formats:
   - Open Table Format (Delta, Iceberg, Huidi)
   - Opensource format (Parquet, Avro)
   - Datawarehouse (Dremio, Druide, Snowflake, Google Big Query)

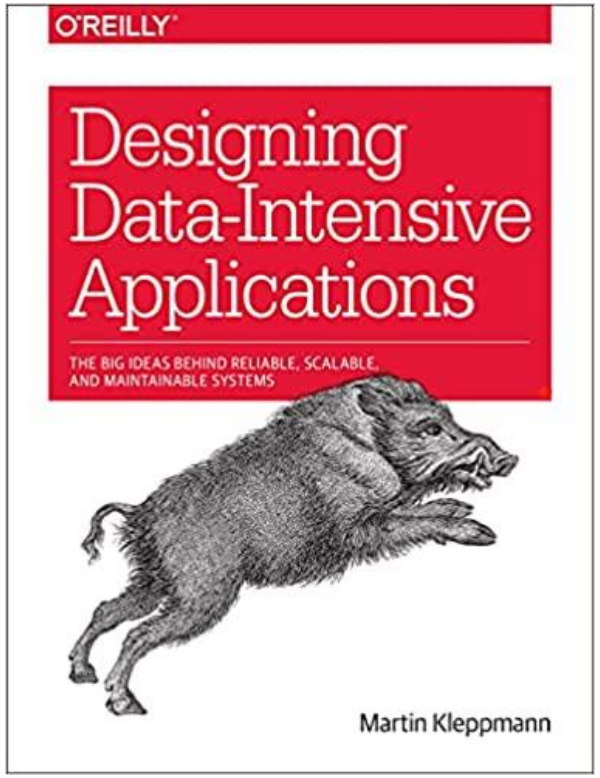
WHAT IS BIG DATA?

"BIG DATA IS LIKE TEENAGE SEX
EVERYONE TALKS ABOUT IT,
NOBODY REALLY KNOWS HOW TO DO IT,
EVERYONE THINKS EVERYONE ELSE IS DOING IT,
SO EVERYONE CLAIMS THEY ARE DOING IT."

DAN ARIELY

Designing Data-Intensive Applications

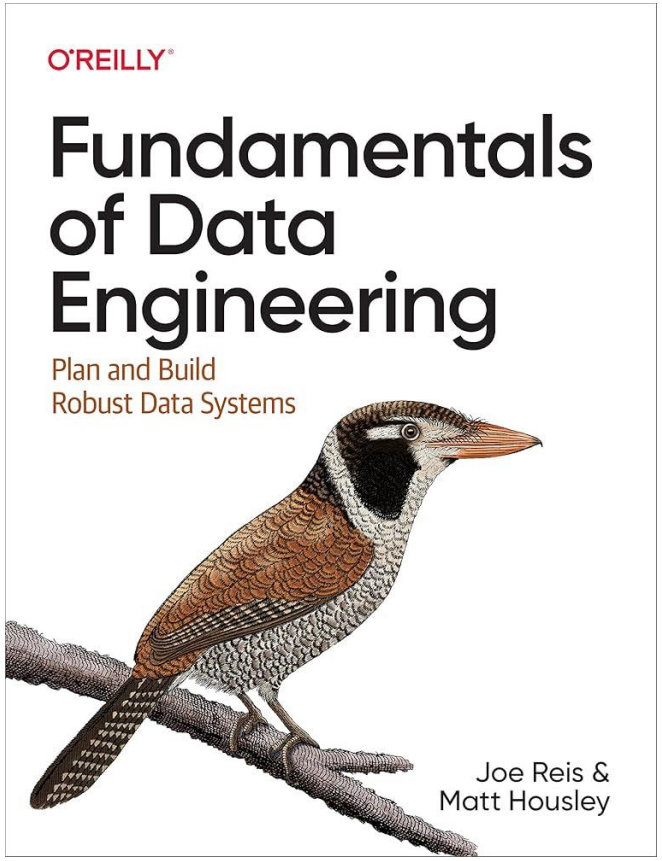






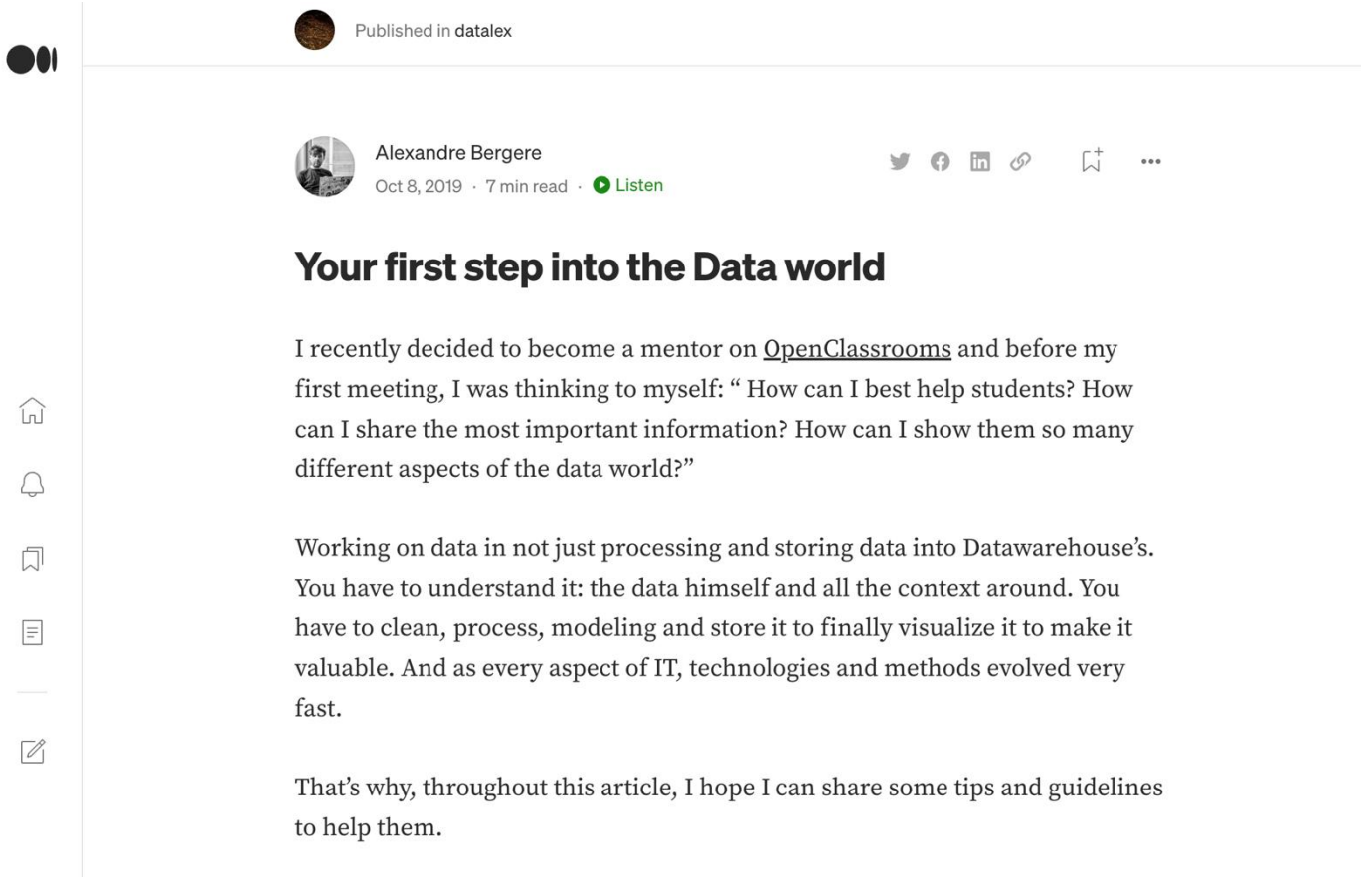
Fundamental of Data Engineering





Free pdf link

Your first step into the Data world



Published in datalex

Alexandre Bergere
Oct 8, 2019 · 7 min read · Listen

Your first step into the Data world

I recently decided to become a mentor on OpenClassrooms and before my first meeting, I was thinking to myself: " How can I best help students? How can I share the most important information? How can I show them so many different aspects of the data world?"

Working on data in not just processing and storing data into Datawarehouse's. You have to understand it: the data himself and all the context around. You have to clean, process, modeling and store it to finally visualize it to make it valuable. And as every aspect of IT, technologies and methods evolved very fast.

That's why, throughout this article, I hope I can share some tips and guidelines to help them.

https://medium.com/datalex/your-first-step-into-the-data-world-b74e24c89d10